
Standalone Chip-on-Board Test Board

Bring-up Plan

Motherboard + interchangeable card: [bonded-die card](#) or [K26 FPGA card](#)

nanoSoC Ethernet chiplet · TSMC 65 nm LP · submitted candidate **rc4-20260829**

Design	nanoSoC Ethernet chiplet, candidate rc4-20260829 (submitted to Europractice / imec on 1 September 2026)
Date	2026-09-30
Status	Draft for review
Audience	The lab's bring-up engineers and the PCB designer. Hardware- and firmware-literate; familiar with the project, not assumed to know every erratum.
Scope	A bring-up motherboard with one card slot, and interchangeable cards for that slot: the die card (one bare die wire-bonded, chip-on-board, to a small card) and the K26 FPGA card (a Kria K26I SOM running the chiplet emulator), used to bring the motherboard up first; optionally a packaged-die card. The die is brought up without the compute chiplet. The TideLink die-to-die interconnect is out of scope : the die card only parks its pins safely.
Repository state read	SoC-Labs/NanoSoC-Ethernet-Chiplet at 914c9e4 (branch feat/padring-boundary-scan); nanosoc-multicore-system at 2adcd34, whose boot code, include files and RTL are identical to the rc4 pin f60449b.

How to read this document. Section 1 is the one-page summary. Section 4 is the PCB designer's section; Section 3 defines the slot between motherboard and card. Sections 5 and 6 are the bench procedure: Phase M (motherboard with the K26 FPGA card), then phases P0–P15 on the die card, each ending in a gate that must pass before the next phase starts.

Source references are printed small and grey, e.g. REG: 170-196. They are *path:line* references into the repository; the short aliases (REG, PROC, BSC, ...) are expanded in Appendix E.

NOT FOUND means the fact was searched for in the repository and is not recorded there. Every NOT FOUND item is carried as an open question in Section 10.

Vendor data. This file lives in a public repository. TSMC IO-library parameters (pull resistances, input thresholds, drive currents, EM limits) are therefore not reproduced; read them from the site-licensed IO databook ([ASIC/tech_wrappers/tsmc65/README.md:21-24](#)).

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1 Summary

What the board proves. The board is a **motherboard with one card slot** that takes **interchangeable cards**: the **die card** carrying one wire-bonded rc4 die, or the **K26 FPGA card** running the chiplet emulator. The die goes from first power to Ethernet traffic with no compute chiplet attached. The phases in Section 6 establish, in order:

Phases	What passing proves
M	The motherboard works before any die is at risk: rails, clocks, flash programming path, PHY link, HOSTIO header and host-MCU scripting, proven with the K26 FPGA card.
P0–P1	The die card is bonded without shorts or opens. Supplies, clock and reset behave. Current draw is plausible.
P2	The pad ring and the boundary-scan TAP work, independently of the core.
P2b	New (B-1). With the DNP loopback footprints fitted: TL pad continuity, then one die trains the shipped D2D configuration against itself.
P3–P7	The HOSTIO4 host link works. Both boot ROMs ran. Clock, PHC, registers and all five RAMs respond.
P8	Whether SWD attaches on a die. SWD has answered on FPGA but never on a die: DPIDR 0x6BA02477 on the HAPS-SX eth-chiplet build (2026-09-09) and on the KR260 eth image (2026-10-02), and the silicon eth ROM read through AP0 on the HAPS-SX BLD-4b mirror (2026-09-30).
P9–P11	The QSPI flash path boots both CPUs. This is the critical path.
P12–P14	On-chip peripherals; Ethernet MAC + RMII PHY to a host PC (ping, UDP echo, throughput); PHC and PTP.
P15	Voltage and frequency shmoo: 50 to 100 MHz, VDD 1.00 to 1.20 V.

Not testable on a standalone board: a two-die TideLink link, and UART, PPS and GPIO (none of them has a pad). With the B-1 loopback footprints fitted, one die can train the shipped D2D configuration against itself (P2b).

The critical path is flash provisioning.

- CPU0 (the network core) is the only core that reaches the Ethernet MAC. CPU1 and the HOSTIO master do not decode 0x40000000 REG:210-217, 259.
- CPU0 always starts in its mask ROM. The ROM copies CPU0's image from QSPI flash into IMEM and halts if the CRC fails ROM0:209-250.
- HOSTIO and CPU1 cannot redirect CPU0. The eth REMAP register at 0x50002000 is decoded only by the debug access port (DAP) REG:259.
- The DAP's only port is SWD. The register says SWD never returned a DPIDR REG:215-216; that is out of date. SWD has answered on FPGA but never on a die: DPIDR 0x6BA02477 on the HAPS-SX eth-chiplet build (2026-09-09) and on the KR260 eth image (2026-10-02), and the silicon eth ROM read through AP0 on the HAPS-SX BLD-4b mirror (2026-09-30). On a die it is still unproven, so flash stays the plan of record.

So the board must let the flash be programmed off-board, through a mux and a programmer header, or be replaced by an SPI-NOR emulator. Every phase from P11 onwards depends on this.

Five things that must exist before first power-on.

1. **A motherboard that passed Phase M with the K26 FPGA card**, carrying: a CLK oscillator with an output-enable (OE) pin, run at 50 MHz; a reset supervisor on NRST; an adjustable, current-monitored core rail; the IOACK pull-up; the QSPI_IO1 pull-down; CARD_ID and CARD_READY gating; scope points. **A die card** with TEST and SE held low and TL_RX/TL_CLK_RX tied down.
2. **A route to CPU0 code that does not need SWD.** A QSPI flash behind a mux with an off-board programmer header, plus an emulator header. The part must accept the ROM's commands 0x0B, 0x06, 0x02, 0x05 and 0x98.
3. **Pre-built, checked flash images.** Start with a TABLE0-only image, packed with `flash_pack.py` and passed by `check_flash.py`.
4. **The HOSTIO host stack, verified on the bench before the die is powered.** This means the status-nibble fix, the resync watchdog, capture to file, and `--target asic_tsmc65`.
5. **A first-power script and record sheet.** They cover the diode checks, current limits, NRST release with CLK stopped, and an address blocklist enforced in the tools.

First action: email Zepler Cleanrooms (zepler.cleanrooms@soton.ac.uk) and settle the bonder, wire type (Au ball or Al wedge), stage size and card-thickness limit, finger size and pitch, and fiducials. That choice fixes the *die card*'s finish (ENIG + gold edge fingers for Al; ENEPIG for Au) and its chip-on-board footprint, so it gates the card layout. The motherboard can be laid out in parallel.

At the same time, ask imec for the sawn die size, die thickness and backside connection, a PadShorts/ERC run on the rc4 GDS, and the POC and power-sequencing note (Section 10).

2 Die and interface

2.1 Die facts

Item	Value	Source
Candidate	rc4-20260829. Submitted 1 Sep 2026. GDS md5 6b0833c4...	REG:6-9
Process, IO library	TSMC 65 nm LP. IO library tphn65lpgv2od3_sl: staggered, 2.5 V devices, 3.3 V-capable	ASIC/genus-innovus/scripts/nanosoc_eth_chiplet_pads.mmmc:26
Die size	Design frame 1600 × 2000 μm . With the imec seal ring 1640 × 2040 μm ; add 20 μm to every design-frame coordinate. IO box 135–1465 × 135–1865; core 205–1395 × 205–1795	FP:12-14 ; docs/plans/TAPEOUT_PUNCHLIST_2026-08-24.md:24-28
Sawn size, scribe, thickness, backside	NOT FOUND . Seal-ring width was asked of the broker and never answered	docs/tapeout/55-imec-preliminary-gds-submission-checklist.md:488
Pads	82 IO cells, 82 bond pads (48 signal + 34 supply) and 4 corner cells. Every IO cell has a bond pad; bond all 82	PNR:1316477-1316558; CSV
Bond-pad rows	Staggered, two rows. Outer row PAD70GU_SL (42 pads), inner row PAD70NU_SL (40 pads)	CSV
Pitch	Alternating 80 μm (160 μm same-row) on the top, right and bottom edges. Alternating 67 μm (134 μm same-row) on the left (D2D) edge	CSV; docs/tapeout/60-waiver-document-rclv3r4.md:158-165
Bond opening	58 μm along the edge × 66 μm deep. Opening centre 36.8 μm (outer) or 138.8 μm (inner) from the design-frame edge; the rows are 102 μm apart. Measured from imec's rc4 BND marker GDS	REG:500-502; research report 01
Supplies	One power domain; no analogue or PLL rail. VDD ×6 and VSS ×4 on the top and bottom edges only. VDDIO ×12 (one of them PVDD2P0C_G) and VSSIO ×12, three of each per side	PADS:240-270; PNR:1321714-1321743
VDDIO level	3.3 V, decided (B-2) . rc4 STA and the UPF used 3.3 V IO; two docs say 2.5 V, but those views were never timed	UPF:110-112; docs/tapeout/10-tapeout-submission.md:163
Clocks	CLK: no PLL, no divider, SYS_HCLK = SYS_FCLK; also aliased to rtc_clk and user_ref_clk. RMII_REF_CLK: 50 MHz input. SWDCK constrained to 25 MHz	PRMU:87-93; SDC:12-14
Reset	NRST is the only reset pad. There is no on-die POR cell	PRMU:133-157
Cores	Two Cortex-M0+. CPU1 (chip core) runs from power-on and owns the fabric clock and reset. CPU0 (network core) is held until CPU1 sets 0x29000000 bit 2	PLAN:332-346
Host ports	SWD (2 pins) and HOSTIO4/ADP (7 pins) only. No UART, GPIO, PPS or CPU-visible I ² C pad	PROC:131-134; CB:137-140
Boundary scan	IDCODE 0x10001001, IR length 4, 76-cell boundary register. TAP shares SWD and HOSTIO[0:1] pins; SE is TRST_n	PTAB; BSC:77-95
Boot ROMs	stage0_bootrom_chip_core (CPU1) and stage0_bootrom (CPU0). Both verified bit-exact against the rc4 GDS	REG:535-536
Timing sign-off	100 MHz target. Setup +15 ps at SS/1.08 V/125 °C. Hold checked at FF corners only. Trust about 95 MHz on slow parts; keep VDD ≤ 1.2 V	REG:350-353, 445-456

2.2 Pad-ring map

Figure 1 is drawn from the as-built pad CSV. Each coloured square is a bond opening at its measured position and size. The script that draws it corrects one CSV defect: row 9 (BuPAD_HOST_IO_6) says edge=top, but the pad is on the **right** edge; its IO cell and its x coordinate both put it there.

2.3 Pads by function

“On-die pull” is the rc4 netlist state of each pad’s pull enable (REN, active-low), traced through the buffer chains. It is not just the source default PTAB; PNR:1321639-1322799. Where it says **none**, the board must bias the pin. Section 3.7 says which of these parts sit on the die card and which on the motherboard.

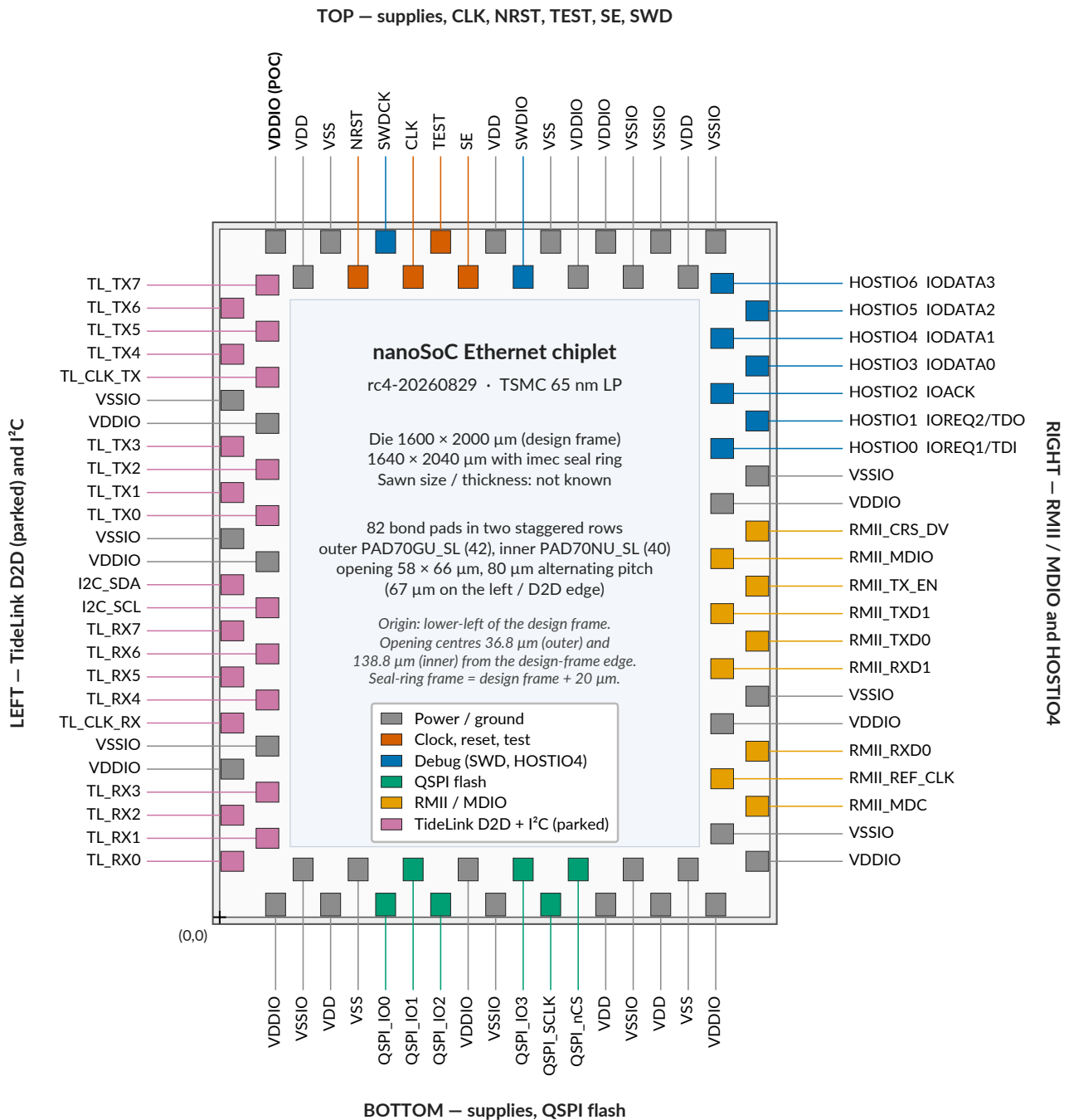


Figure 1. Pad-ring map of the rc4 die, design frame, viewed from the top (pad side), origin lower-left. The squares are bond openings drawn to scale (58 × 66 μm) in their staggered outer and inner rows. The bold label is the single POC supply pad, VDDIO_T_0. Source: csv, drawn by figures/gen_figs.py.

Pad(s)	Direction	Cell	On-die pull	Required board handling	Source
Clock, reset, test					
CLK	in	PDDW04DGZ (non-Schmitt)	none	CMOS oscillator with OE, 50 MHz for boot (B2). Series 22–33 Ω at the source. A weak pull-down holds the net low while OE disables the oscillator. Max input transition 1.2 ns. Test point	PADS:551; SDC:66-67
NRST	in, active-low	PDDW04DGZ	none	Push-pull supervisor output. Button and host MCU drive the supervisor's manual-reset (MR) input. Must never float. Edge ≤ 5 ns. Test point	PADS:567; SDC:71-72
TEST	in	PDDW04DGZ	pull-down	Tie low: 0 Ω or 10 k Ω to VSS, plus a test point. High bypasses the reset and HOSTIO synchronisers	PADS:559; PROC:204-220
SE	in	PDDW04DGZ	pull-down	10 k Ω pull-down plus a jumper to VDDIO; also the JTAG header's TRST_n. SE=1 enables the boundary-scan TAP and turns HOSTIO[0:1] into TDI/TDO	PADS:542; BSC:87-95
Debug: SWD (TCK/TMS in boundary scan) and HOSTIO4					
SWDCK	in	PDDW04DGZ	pull-down	SWD header SWCLK, 22–33 Ω series. Isolation link to the JTAG header (TCK)	PADS:584
SWDIO	bidir	PDUW08DGZ	pull-up	SWD header SWDIO, 22–33 Ω series. Isolation link to the JTAG header (TMS)	PADS:576
HOSTIO[0]	bidir: IOREQ1 out; TDI when SE=1	PDDW16DGZ	none	330 Ω series; 10 k Ω idle pull-up; scope test point. Isolation link to the JTAG header	PADS:643-697; PROC:163-188
HOSTIO[1]	bidir: IOREQ2 out; TDO when SE=1	PDDW16DGZ	none	330 Ω series; 10 k Ω idle pull-up. Isolation link to the JTAG header	same
HOSTIO[2]	bidir: IOACK in, async	PDDW16DGZ	none	Pull-up mandatory , about 5 k Ω , on the die side. Low or floating wedges the HOSTIO FSM in TXC1. 330 Ω series; scope test point	PROC:221-246
HOSTIO[3:6]	bidir: IODATA[0:3]	PDDW16DGZ	none	330 Ω series; 10 k Ω idle pull-ups	PROC:221-246
QSPI flash					
QSPI_SCLK	out, always driven	PDDW16DGZ	n/a	22–33 Ω series at the die; through the mux to the flash; test point	PADS:593-640
QSPI_nCS	out, always driven	PDDW16DGZ	n/a	10 k Ω pull-up on the flash side, which keeps the flash deselected during ramps and mux switching	same
QSPI_I00	bidir	PDDW16DGZ	none	Weak pull-up on the die side (the die drives it high at idle in single-SPI mode)	qspi_controller. sv:673-676
QSPI_I01	bidir	PDDW16DGZ	none	Pull-down, 10 kΩ, on the die side. With blank or absent flash the boot then parks in 1.0 ms, not after a 27.5 s busy-bit poll at 50 MHz (rc4-RTL sim, B-6). Test point	REG:242-252; docs/chip-reference-manual/review/closure/item08-bringup-dryrun.md
QSPI_I02/I03	bidir	PDDW16DGZ	none	Pull-ups on both sides of the mux: they are the flash's WP# and HOLD#	PADS:593-640
RMII and MDIO					
RMII_REF_CLK	in	PDDW04DGZ	none	50 MHz from an oscillator through a fan-out buffer, length-matched to the PHY. DNP 0 Ω option to take the PHY's REF_CLK output instead. Weak pull-down; test point	PADS:700-770; SDC:13
RMII_RXD0/1, CRS_DV	in	PDDW04DGZ	pull-down	Direct to the PHY, short traces	same
RMII_TXD0/1, TX_EN	out, always driven	PDDW16DGZ	n/a	22–33 Ω series	same
RMII_MDC	out, always driven	PDDW16DGZ	n/a	22–33 Ω series; test point	same
RMII_MDIO	bidir	PDDW16DGZ	none	External pull-up, about 1.5 kΩ. An inadequate pull-up caused the KR260 MDIO fault. Test point	PROC:881
TideLink D2D and I²C sideband (parked; out of test scope)					
TL_RX[7:0], TL_CLK_RX	in	PDDW16DGZ	none	Bond them, then pull each to VSSIO through 10 k Ω . Left unbonded they float on the die	PADS:846-917

Pad(s)	Direction	Cell	On-die pull	Required board handling	Source
TL_TX[7:0], TL_CLK_TX	out, always driven	PDDW16DGZ	n/a	Cannot be tri-stated, even in EXTEST. Never tie to a rail. Bond them (B-1); route to test points and to DNP 0 Ω TX[n]→RX[n] loopback footprints, used for boundary-scan continuity and P2b	PADS:774-845; BSC: 406-412, 469-476
I2C_SCL/SDA	open-drain (input tied 0)	PDUW16DGZ	pull-up (weak)	2.2–4.7 kΩ pull-ups to VDDIO, plus test points. Do not connect them to the board-management I ² C bus	PADS:918-948
Supplies (bond all 34)					
VDD × 6	core supply	PVDD1DGZ_G	–	1.20 V nominal; top and bottom edges only. Decouple at the fingers, outside the glob-top keep-out	UPF:107-109
VSS × 4	core ground	PVSS1DGZ_G	–	Board ground plane; top and bottom edges only	PADS:256-270
VDDIO × 12	IO supply	11 × PVDD2DGZ_G + 1 × PVDD2P0C_G	–	3.3 V (open question 1). The POC pad is VDDIO_T_0, first pad on the top-left. Three pads per side; decouple on every side	PNR: 1321714-1321725
VSSIO × 12	IO ground	PVSS2DGZ_G	–	Board ground plane; three per side	PNR: 1321732-1321743

2.4 Pad behaviour the board must respect

- **15 outputs are always driven** (OEN tied low): QSPI_SCLK, QSPI_nCS, RMII_TXD0/1, RMII_TX_EN, RMII_MDC, TL_TX[7:0], TL_CLK_TX. No instruction can release them, boundary-scan EXTEST included BSC:400-412.
- **No on-die pull** on CLK, NRST, RMII_REF_CLK, RMII_MDI0, QSPI_IO[3:0], HOSTIO[6:0], TL_RX[7:0] or TL_CLK_RX. The board biases every one of them REG:200-203.
- **No straps.** role_strap, mask_hs_bypass and apb_debug_unlock are tied 0 on the die; nego_priority is tied 0x0001 build/chip/rtl/nanosoc_eth_chiplet_chip.v:113-116.
- **Not pads** (despite older docs): UARTs, PL022 SPI, JTAG-DP TDI/TDO/nTRST, scan pins, 1PPS. docs/design/PIN_MAP.md §2 and §5 are stale on this point.
- **Electrical limits** (input thresholds, maximum input voltage, drive current, pull resistance) are in the TSMC IO library databook (site PDK, staggered 2.5 V IO library); they are not reproduced here.

3 Slot interface (contract v1)

The board is split in two: a **motherboard** with one card slot, and **interchangeable cards** that plug into it. The user decided (2026-09-30) that the bonded-die card and an FPGA SOM card must be interchangeable in the *same* slot, so the FPGA sits on a plug-in card, not on the motherboard. This section is **contract v1** (research report 06), which supersedes v0. It fixes what crosses the slot so the motherboard and the cards can be designed in parallel. The full pin-by-pin assignment is the next deliverable (Section 3.8).

3.1 Card types

CARD_ID	Card	Rails the host MCU enables	Role
000	Die card	VDDIO, then VDD_CORE (sequence S1)	The rc4 die wire-bonded (chip-on-board) onto a small card. Phases P0–P15
001	K26 FPGA card	VDDIO and VIN_AUX; never VDD_CORE	AMD Kria K26I SOM (5M-K26-XCL2GI) on a slot-format carrier. Runs the eth-chiplet FPGA build (the emulator). Brings the motherboard up in Phase M and stays as the golden reference. Carrier details: FPGA_DAUGHTER_CARD_OPTIONS.pdf (same directory)
010	Packaged-die card (optional)	VDDIO, then VDD_CORE	A Europractice-packaged die (QFN or ceramic) on a card: the packaging hedge on the same motherboard
111	Slot empty	none	Pull-ups read all ones
other	reserved	none	Treat as a fault

Why the K261. It is the same XCK26 family as the proven KR260 build (full chiplet about 58k LUT, 51%), its 69 HDIO banks at 3.3 V cover all 48 slot signals including TideLink, and the nanoSoC FPGA Toolkit already supports zynqmp. On 2026-09-30 Mouser UK had 112 K261 in stock (£439.21) against none for the K26C (£353.26, 969 on order); the two differ only in temperature grade and shock/vibration qualification (DS987 v1.2), but the Vivado part becomes xck26-sfvc784-2LV-i. The K24 was rejected: 23 HDIO at 3.3 V (DS985), level shifters for 1.8 V HPIO, and about 82% of its LUTs. Fallback: Trenz TE0712 (Artix-7 200T), which had zero stock and needs a toolkit bin_style extension.

3.2 Why a die card, and why not flip-chip

The first draft bonded the die straight onto the motherboard. A flip-chip die on a card was also studied (research report 05) and rejected:

- **rc4 is a wire-bond die.** imec ran the Wirebond BEOL option; IM.POLYIMIDE.1 fires on all 82 pads (“if bumping is required ... polyimide may be a must have”), and the stream has no UBM geometry [docs/tapeout/60-waiver-document-rc1v3r4.md:108-115,149-151](#).
- **The substrate cannot escape the ring.** With 60 µm landing pads, one trace between outer-row pads needs about 33/33 µm line/space, and 25/25 µm on the 134 µm-pitch left edge. Standard PCB is 100 µm and HDI 50–100 µm; only UHDI, fine-line flex or thin film fit, all quote-only.
- **Joints cannot be inspected or reworked.** rc4 never ran ERC or PadShorts, so on a flip-chip card a dead die cannot be told from a bad bump. Au studs from 25 µm wire also overhang the 58 × 66 µm openings.

	(a) Flip-chip on a card	(b) COB on a card (chosen)	(c) Europractice package	(d) COB on the motherboard (first draft)
Feasible here	In principle only: ~25 µm L/S substrate, a bump process on a wire-bond pad stack, tuning on real dies	Yes: standard 100 µm-rule PCB, same bond rules as (d)	Yes, if the die is ≤ 300 µm (QFN) or ≤ 600 µm (ceramic); thickness unknown	Yes
Risk	High: substrate, pad damage, hidden opens, no rework	Low-medium: staggered-row loop plan	Low. Minimum order 10; long wires in a 12 × 12 QFN	Low-medium: a large board may not fit the bonder
Cost, 3–10 units	Substrate quote-only, plus bump development	Standard 4-layer cards (not quoted) + one motherboard	€485 + 10 × €83–229 (QFN) $\approx$ €1.3–2.8k; sockets extra	N motherboards, each with a bondable finish
Die swap / reuse	Swap the card; die not recoverable	Swap the card; motherboard reused	Yes, with a socket (card 010)	No: one die per motherboard
Debug	X-ray and board test points only	Probe fingers before glob top; lift a wire to isolate a pad	Package pins only	Same as (b)
Motherboard impact	Add a connector	Add a connector; any fab and finish	Socket or connector	None

Sources (research report 05): arXiv 1112.1037 (Au stud bumps on single dies); Würth HDI microvia poster; Eurocircuits HDI pool; PCBWay capabilities; Europractice packaging rules and service page; Technion TPT HB16 page [search snippet]; [docs/tapeout/60-waiver-document-rc1v3r4.md:108-115,149-172; REG:500-502](#).

Why the card is kept even without flip-chip: a small card fits the bonder’s stage (an HB16-class bonder takes samples up to 100 × 150 mm [search snippet]); only the die card needs a bondable finish, so the motherboard can come from any fab; a bad die or bond scraps a small card, not a populated motherboard; the same slot takes the K26 FPGA card, so the motherboard is brought up before dies arrive; and a later two-slot motherboard could host a TideLink die pair.

3.3 Connector and mechanics

- **Samtec HSEC8** card edge: 0.8 mm pitch, 1.60 mm card (the -01 option), **120 positions**. The pin budget below uses about 107–109, leaving about 11–13 spare.
- Alternative: Samtec Q Strip mezzanine (0.5 mm pitch, ground plane, no hard-gold fingers).
- **Die-card finish:** ENIG + gold edge fingers for Al wedge bonding (Eurocircuits offers ENIG with a gold-edge-connector option). ENEPIG (needed for Au ball) plus hard gold on one board is **unconfirmed**.
- Ground fingers are longer, so ground mates first. Same keying for every card type.
- **Mechanical envelope:** the die card is small. The K26 FPGA card is up to about 100 × 90 mm: the SOM (77 × 60 mm) plus heatsink and fan. The motherboard keeps a keep-out volume over the slot and provides a **mechanical support** for the heavier FPGA card (card guide, or standoff and bracket).
- At ≤ 100 MHz the connector’s bandwidth is not the constraint. What matters is enough ground returns next

to the 15 always-driven 16 mA outputs, and series resistors and decoupling on every card.

3.4 Pin groups

All 48 signals cross the slot one-to-one, named as the die pads, as 3.3 V LVCMOS. [v1] marks what changed from v0.

Group	Pins	Count	Notes
Control and clock	CLK, NRST, TEST, SE	4	TEST tied low on every card; SE pulled down on every card, motherboard jumper can pull it to VDDIO
SWD	SWDCK, SWDIO	2	Also boundary-scan TCK/TMS
HOSTIO4	HOSTIO[6:0]	7	HOSTIO[0]/[1] are TDI/TDO when SE=1
RMII	RMII_REF_CLK, RMII_TXD[1:0], RMII_TX_EN, RMII_RXD[1:0], RMII_CRS_DV, RMII_MDC, RMII_MDIO	9	
QSPI	QSPI_SCLK, QSPI_nCS, QSPI_IO[3:0]	6	
TideLink	TL_TX[7:0], TL_CLK_TX, TL_RX[7:0], TL_CLK_RX	18	Routed so a two-slot D2D motherboard stays possible. Parked on the card; no load on the single-slot motherboard
I ² C sideband	I2C_SCL, I2C_SDA	2	TideLink-owned; pulled up on every card
Signals		48	
VDD_CORE	motherboard-adjustable 1.00–1.25 V	~6	Only for card types 000 and 010. [v1] Optional DNP dummy load on the motherboard, to test the core supply with the FPGA card fitted
VDD_CORE_SENSE_P/N	Kelvin sense from the card	2	Regulation or measurement at the die, not at the motherboard
VDDIO	3.3 V (2.5 V option)	~6	[v1] On for every card type. The FPGA card powers its slot-side bus switches from it, so the slot nets always sit at the same logic levels
VDDIO_SENSE_P/N	Kelvin sense from the card	2	
VIN_AUX	5 V, [v1] ≥ 4 A continuous	≥ 6	For the K26 card (up to about 4 A at 5 V). HSEC8 about 2.8 A/pin is a search-snippet figure: derate. Through an eFuse with an INA228. The die card leaves it unconnected
GND	at least one per 2 signals, longer fingers	≥ 26	Place returns next to the 15 always-driven outputs
CARD_ID[2:0]	strapped on the card (GND or open), pulled up on the motherboard	3	000 die, 001 FPGA, 010 packaged die, 111 empty
CARD_PRESENT_n	one at each end of the connector	2	Both low = fully mated
CARD_READY	[v1] card → motherboard, open-drain, pulled up on the motherboard	1	The FPGA card releases it only when configured <i>and</i> its slot IO is enabled; the die card straps it ready
Card JTAG	[v1] TCK, TMS, TDI, TD0, VREF on spare pins	5	Lets the motherboard reach the FPGA card's JTAG chain; the FPGA card also keeps its own USB-JTAG/UART
CARD_I2C_SCL/SDA (optional)	ID EEPROM or temperature sensor on the card	2	Separate from the die's I ² C
Total		~107–109	120-position HSEC8 leaves ~11–13 spare

3.5 Power-enable rule

1. With every slot rail off, the host MCU checks that both CARD_PRESENT_n are low. If not, no rail is enabled.
2. It reads CARD_ID[2:0]. The pull-ups must run from the MCU's own always-on supply, not from a slot rail, so the read works before any rail is on.
3. 000 or 010: enable VDDIO, then VDD_CORE, by sequence S1 (Section 4.5). 001: enable VDDIO and VIN_AUX (through its eFuse); **never VDD_CORE**. 111 or anything else: nothing.
4. The host waits for CARD_READY before it releases NRST or starts any test. On the die card it is strapped ready; on the FPGA card it follows configuration.
5. No hot-plug: insert or remove a card only with all slot rails off. If either CARD_PRESENT_n goes high while powered, the host drops every rail at once.

3.6 Rules every card must meet

- **Same net behaviour as the die.** Every card carries the die-card terminations: TEST low, SE 10 k Ω pull-down, TL_RX[7:0]/TL_CLK_RX 10 k Ω pull-downs, I²C 4.7 k Ω pull-ups, and series resistors on its outputs.
- **The FPGA card isolates its slot IO** with bus switches (for example SN74CB3Q3245, Ioff-rated) until it is configured, so FPGA configuration pull-ups never reach the slot nets.
- **No hot-plug.** The host MCU turns every rail off before a card is inserted or removed.

3.7 What sits where

Die card (000)	K26 FPGA card (001)	Motherboard (everything else)
Die, bond fingers, VSS die-attach pad, glob-top keep-out	K26I SOM on its 2 × 240-pin connectors; power and sequencing from VIN_AUX (UG1091); heatsink and fan	VDD_CORE (TPS62869, 1.00–1.25 V), VDDIO, VIN_AUX with eFuse; INA228 on every rail; current limits; optional VDD_CORE dummy load
VDD and VDDIO decoupling at the die	Bus switches on all 48 slot signals	CLK oscillator 25/50/100 MHz with OE gating; NRST supervisor + button
Terminations: TEST low; SE 10 k Ω down; TL_RX, TL_CLK_RX 10 k Ω down; I ² C 4.7 k Ω up; TL_TX test points	The same terminations	QSPI flash (SST26VF064B-class), TS3A27518E mux, programmer and emulator headers, QSPI pulls (IO1 down; IO2, IO3, nCS up)
Series resistors on die-driven outputs: 22–33 Ω on RMII TXD/TX_EN/MDC and QSPI SCLK/nCS	Series resistors on its outputs	LAN8720A + MagJack; 50 MHz oscillator and buffer to the PHY and the slot REF_CLK; 1.5 k Ω MDIO pull-up
CARD_ID straps 000; CARD_READY strapped ready; optional ID EEPROM	CARD_ID straps 001; CARD_READY driver	HOSTIO4 header (330 Ω series, IOACK pull-up about 5 k Ω , idle pull-ups); SWD 10-pin header; JTAG header with isolation links; SE jumper
VIN_AUX not connected	USB-JTAG/UART; boot from SOM QSPI (optional microSD); status LED	RP2350 host MCU reading CARD_ID, CARD_PRESENT_n and CARD_READY; mechanical support and keep-out for the FPGA card

3.8 Next step: the pin-by-pin assignment

Contract v1 fixes groups and counts, not positions. Before either layout starts, produce the full HSEC8 assignment. It must spread VIN_AUX over at least 6 contacts, place a ground next to each of the 15 always-driven 16 mA outputs, route each sense pair as a pair, put CARD_PRESENT_n at the two ends, keep the Tidelink group contiguous for a later two-slot board, and record the spares. It is open question 19 in Section 10.

Topic	Decision	Status
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4 Board requirements

4.1 Decisions at a glance

Topic	Decision	Status
Architecture	Motherboard with one HSEC8 card slot and interchangeable cards : the die card (COB die) or the K26 FPGA card (Kria K26I SOM on a slot carrier, Phase M and golden reference); optionally a packaged-die card. The host MCU reads CARD_ID before any rail and waits for CARD_READY before NRST (Section 3)	User decision 2026-09-30; slot contract v1
VDDIO	3.3 V (B-2): rc4 STA and UPF used only the 3.3 V IO views; 2.5 V was never timed	Decided 2026-10
Core VDD	On the motherboard: adjustable 1.00–1.25 V in 5 mV steps (TPS62869-class, I ² C), nominal 1.20 V, with Kelvin sense from the card through the slot. Never 1.32 V. INA228 on every rail. Current-limited bench supplies for first power	Decided
CLK	Programmable or selectable oscillator: 50 MHz default and boot clock (B2: with CLK_DIV = 1 the QSPI read fails at 100 MHz), 100 MHz only after firmware has set CLK_DIV ≥ 2, plus a 25 MHz option (every FPGA build runs at 25 MHz). The frequency must be 10 ⁹ / <i>N</i> Hz for an integer <i>N</i> . OE-gateable so NRST can be released with CLK stopped; OE also driven by the host MCU	Decided; part TBD
Reset	Push-pull supervisor plus button on NRST; host MCU also controls NRST. NRST released with CLK stopped (B-4, advised; a CLK-running release is benign)	Decided; part TBD
Ethernet	LAN8720A (FPGA-proven; firmware expects PHY ID 0x0007/0xC0F1) with a MagJack. 50 MHz REF_CLK oscillator through a fan-out buffer to both PHY and die; DNP option for PHY REF_CLK-out mode	Decided
Flash	Exactly SST26VF064B (B-3; the part every simulation and GLS boot models) behind a TS3A27518E mux to an off-board programmer header, plus an SPI-NOR emulator header. No alternate. Series-R footprints and scope points on SCLK/IO0/IO1 at the die (B-5)	Decided 2026-10
Debug	HOSTIO4 7-pin header (primary; RP2350 host). SWD 10-pin 1.27 mm Cortex header. JTAG boundary-scan header isolated from both by links. J-Link BASE (1.2–5 V) as the SWD/JTAG probe	Decided
TideLink	Parked on the die card by default: RX pulled down, TX bonded to test points, DNP 0 Ω TX→RX loopback footprints (B-1), I ² C pull-ups. All 18 TL signals still reach the slot for a later two-slot board. Tools never write ROLE_CFG, except in P2b with the loopback fitted	Decided 2026-10
Assembly	Chip-on-board on the die card at Zepler Cleanrooms (manual Au/Al bonder), 2–3 cards. Card finish ENIG + gold edge fingers (Al wedge). The motherboard needs no bondable finish. Hedge: about 10 Europractice-packaged dies on packaged-die cards	Bonder rules open (Q4)

4.2 Changes from the pre-silicon closure (October 2026)

The pre-silicon closure runs on the exact rc4 RTL and netlist (docs/chip-reference-manual/review/PRE_SILICON_CLOSURE_PLAN.md §1; item files in <docs/chip-reference-manual/review/closure/>) changed six board decisions. The 8–9 October runs (items 6, 6b, 6g, 13, 17) relaxed B-4 to *advised*, fixed the boot clock at 50 MHz and closed EM (P2). All evidence is simulation or STA; none is silicon.

#	Change	Evidence
B-1	Bond TL_TX[7:0]/TL_CLK_TX (closes open question 10) and add DNP 0 Ω footprints TL_TX[n]→TL_RX[n] and TL_CLK_TX→TL_CLK_RX next to the die, length-matched; keep the 10 kΩ pull-downs. New phase P2b .	One rc4 die trains against itself as die_a and as die_b (cal_done 23.7 ms at 50 MHz, 7 FC nodes, 8 lanes, CRC on) and moves 16 peer words byte-exact; without the loopback it never trains docs/chip-reference-manual/review/closure/item03-d2d-loopback.md
B-2	VDDIO = 3.3 V, decided (closes open question 1).	STA used only the 3.3 V IO views; the 2.5 V views were never timed. Also suits SST26VF064B, RP2350, LAN8720
B-3	Fit exactly SST26VF064B , not “SST26-class”; put the same part on the K26 card's slot side. Drop the MX25R6435F alternate (closes open question 6).	Every sim and GLS boot uses the vendor SST26VF064B model; the ROM's 0x98 unlock and boot-counter write are proven only against it
B-4	NRST release with CLK stopped is advised, not mandatory (changed 2026-10-08): keep the OE-gated glitch-free oscillator and MCU-controlled NRST as the clean method; a CLK-running release is not a bring-up blocker. Firmware clears the perf counters and ETC and programs the PHC after boot.	rc4 SS-SDF GLS: removal violations only within about ±1 ns of a CLK edge docs/chip-reference-manual/review/closure/item10-gls.md . STA: 65–69 % of release phases at SS hit removal on some raw-NRST flop, but every such flop is benign (perf counters, ETC, PHC, SysTick phase); no boot-path FSM is on the raw tree docs/chip-reference-manual/review/closure/item06-sta.md §6a
CLK	Oscillator default 50 MHz for boot; P9 and P11 run at 50 MHz ; raise CLK only after firmware has set CLK_DIV ≥ 2.	STA, corrected QSPI constraint: data is captured on the <i>rising</i> SCLK edge, so the read budget is one CLK period. At CLK_DIV = 1 the read fails at 100 MHz (–5.0 ns at SS) and 90.9 MHz on every corner, typical included; passes at 50 MHz (+5.0 ns at SS); zero slack at ≈66.7 MHz (SS) docs/chip-reference-manual/review/closure/item06-sta.md §6c. Basis: recovered SDC (the shipped SDC was deleted)
P2	EM closed; no board change . Keep steady core current below ≈35 mA at 125 °C.	rc4 Voltus with the GLS SAIF at 100 MHz: IR 4.1 mV, EM J/Jmax 0.38, 0 over, 13.4 mA docs/chip-reference-manual/review/closure/item13-ir-em.md
B-5	QSPI series-R footprints and scope points on SCLK, IO0 and IO1 at the die ; keep the 10 ⁹ /N oscillator (25/50/100 MHz).	QSPI is the critical path and B2 has no valid timing number (STA checked the wrong edge); P9/P15 measure the capture margin. Since 10-08 STA gives the numbers (row CLK)
B-6	Keep the QSPI_IO1 pull-down (confirmed).	rc4-RTL dry run: CPU1 parks in 1.0 ms with the pull-down (flash blank or absent); with IO1 high the busy-poll takes 27.5 s at 50 MHz docs/chip-reference-manual/review/closure/item08-bringup-dryrun.md

Four firmware rules from the closure apply to every image used on this board and any later pair board: **the DMA-250 never targets 0x2E000000--0x2FFFFFFF** (erratum D2D-DMA, Section 8); **never clear TideLink's out_prepend_swi_polarity** (RX-POL: it resets to 1, falling-edge RX capture, which accepts compute's TX with ≥ 2.70 ns; rising-edge capture fails hold by 0.85 ns) [docs/chip-reference-manual/review/closure/item06g-d2d-rx-window.md](#); **no image may use flash 0x000000--0x5FFFFF** except TABLE0/TABLE1, the erased counter and the CPU1 slot/golden images at their address-map offsets (FLASH-LAYOUT; XiP/overlay cold code goes at 0x80000, Appendix B) [docs/chip-reference-manual/review/closure/item17-silicon-rom-envs.md](#); and **on any peer bus error or peer timeout, POR both dies and re-train** (a dropped link-clock edge wedges the link with every FCSM still reading 4; detect a wedge with a canary read of a known peer word, not FCSM state) [docs/chip-reference-manual/review/closure/item07-d2d-traffic.md](#). Expected values for P3–P11 come from golden_asic_reference.json, generated from the rc4-RTL dry run [docs/chip-reference-manual/review/closure/item08-bringup-dryrun.md](#).

4.3 Block diagram

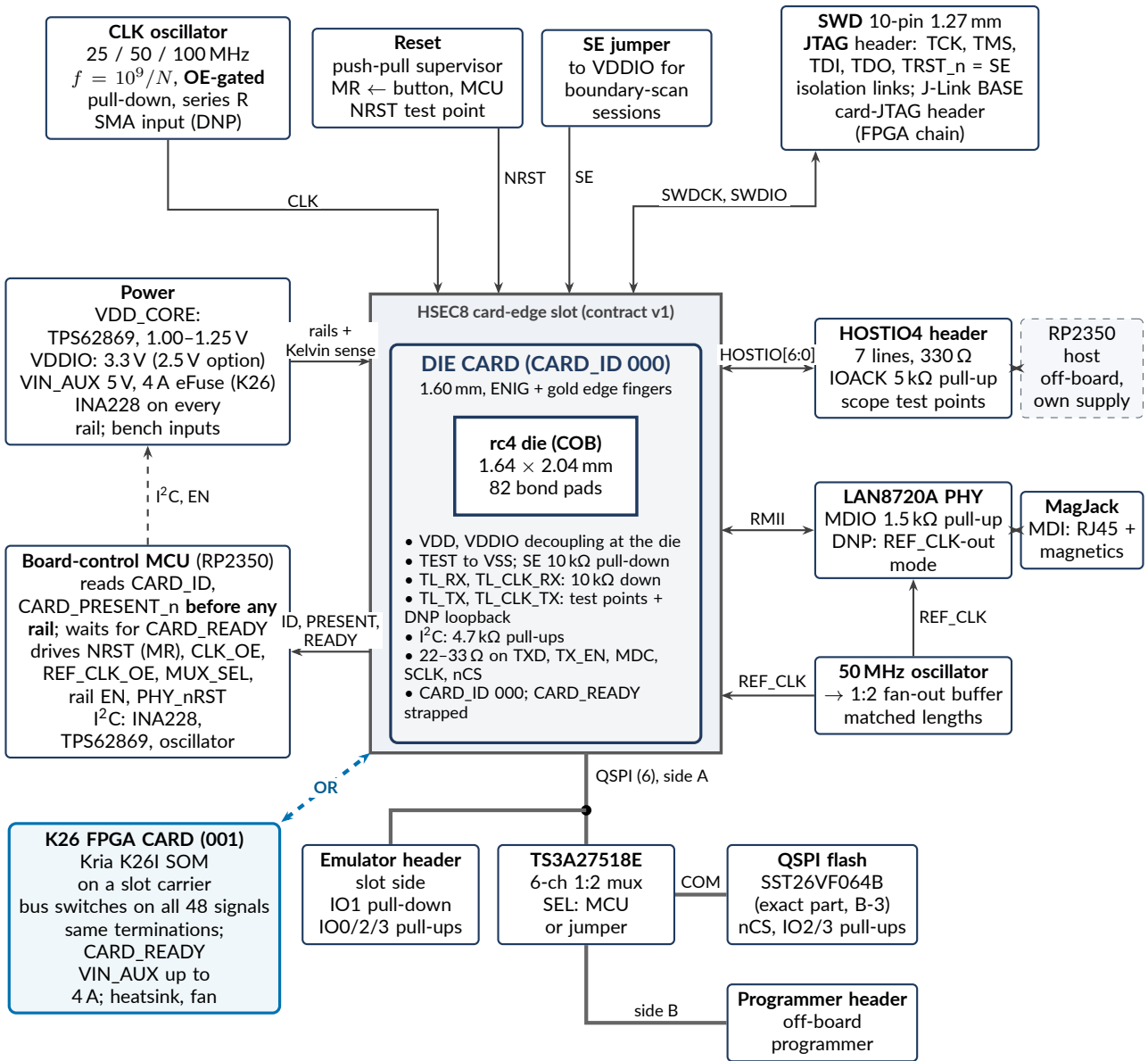


Figure 2. Board block diagram. The slot (grey frame) takes **one of two interchangeable cards**: the die card (shown fitted, centre) or the K26 FPGA card (bottom left); an optional packaged-die card (010) is a third. All carry the same terminations and use the same pins. Every other solid box is on the motherboard, which also provides a mechanical support for the heavier FPGA card. Dashed boxes are off-board. Arrows end at the slot because every signal crosses it one-to-one by pad name (Section 3). Thick lines are multi-signal buses; “DNP” means placed but not fitted.

4.4 Per-block requirements

Parts named in the “Suggested part” column come from the parts survey; anything marked TBD still needs a part choice. Resistor values the research did not fix are engineering starting points, marked “(start)”. **Where:** MB = motherboard, Card = die card. The K26 FPGA card’s contents are in the partition table (Section 3.7).

Block	Where	Requirement	Suggested part	Source
Die site (COB)	Card	Bond all 82 pads. VSS die-attach pad (substrate connection to confirm). One solder-mask opening over the bond area. Glob-top keep-out ring with no parts inside it. Fingers per Section 4.10. Decoupling for VDD and VDDIO on every die side, just outside the keep-out	Die, epoxy and glob top chosen by the bonder	Section 4.10
Card edge	Card	HSEC8 card edge, 1.60 mm card, ENIG + gold edge fingers, longer ground fingers. CARD_ID straps 000; CARD_READY strapped ready; VIN_AUX and card-JTAG pins not connected. Optional ID EEPROM on CARD_I2C	Samtec HSEC8 mating edge	Section 3

Block	Where	Requirement	Suggested part	Source
Slot, card detect, mechanics	MB	HSEC8 socket, 120 positions. CARD_ID[2:0], both CARD_PRESENT_n and CARD_READY pulled up from the MCU's always-on supply (Section 3.5). Keep-out volume over the slot and a card guide or standoff/bracket for the K26 card (up to about 100 × 90 mm with heatsink and fan). Card-JTAG pins to a header	Samtec HSEC8, 0.8 mm	research report 06
Core supply VDD_CORE	MB	1.00–1.25 V adjustable in 5 mV steps over I ² C; nominal 1.20 V. The host tool clamps the setpoint at ≤ 1.25 V, because the regulator's own range reaches 1.675 V. Choose a variant that powers up at ≤ 1.20 V. Enabled only for CARD_ID 000/010; optional DNP dummy load to test it with the K26 card fitted. Kelvin sense from the card (VDD_CORE_SENSE_P/N). A jumper selects the on-board regulator or a bench-supply input. INA228 on VDD_DIE	TPS62869; INA228	UPF:107-109; REG:403-405, 484-485
IO supply VDDIO	MB	3.3 V, on for every card type (the K26 card powers its slot bus switches from it). A 2.5 V fallback must need only a resistor or regulator change. One regulator with two measured branches: VDDIO_DIE (to the slot) and VDDIO_BOARD (flash, PHY IO, oscillators, pull-ups). This keeps them at one voltage and one sequence. Kelvin sense from the card (VDDIO_SENSE_P/N). INA228 on each branch; jumper to a bench-supply input	TPS7A94 (resistor-set); INA228	research report 03 §C3
VIN_AUX	MB	5 V to the slot for the K26 card only (CARD_ID 001): ≥ 4 A continuous on ≥ 6 contacts, through an eFuse, with an INA228	eFuse TBD; INA228	research report 06
PHY analogue supply	MB	From a fixed 3.3 V rail, so that only the PHY's VDDIO follows the die VDDIO. Check the PHY datasheet	LAN8720A datasheet	–
CLK	MB	CMOS, full swing at VDDIO, max transition 1.2 ns at the die, through the connector. 25 MHz (FPGA builds), 50 MHz for the die's boot (default; B2), 100 MHz only after firmware sets CLK_DIV ≥ 2; every setting must be 10 ⁹ /N Hz. OE driven by the host MCU, with a jumper override. Output start after OE must be glitch-free. Pull-down 100 kΩ (start) on the slot side so CLK rests low while the oscillator output is disabled. Series 22–33 Ω at the source. DNP SMA for an external generator (shmoo). Test point next to the slot	TBD: programmable oscillator, or 25, 50 and 100 MHz parts with OE	SDC:12, 66-67; PROC:190-203
Reset	MB	Push-pull, active-low supervisor output on NRST, monitoring VDDIO (and VDD if dual-input). Button and host-MCU open-drain line into MR. Edge ≤ 5 ns. Test point	TBD (push-pull output, MR input)	SDC:71-72; REG:359-377
Straps	Card + MB	Every card: TEST 0 Ω (or 10 kΩ) to VSS plus a test point; SE 10 kΩ pull-down. MB: SE jumper to VDDIO and the JTAG header's TRST_n pin	–	PROC:204-220
QSPI flash	MB	Accepts 0x0B (fast read, 8 dummy clocks), 0x06, 0x02, 0x05 and ULBPR 0x98. 3-byte addressing; keep the image under 4 MB. The mux common side goes to the flash, side A to the slot, side B to a programmer header; SEL from the host MCU plus a jumper, default A. Emulator header on the slot side. Slot side: IO1 pull-down 10 kΩ, IO0/IO2/IO3 pull-ups 47 kΩ (start). Flash side: nCS, IO2, IO3 pull-ups 10 kΩ. SCLK/nCS series resistors sit on the card; series-R footprints and scope points on SCLK, IO0 and IO1 at the die (B-5)	SST26VF064B (3.3 V), exact part (B-3); TS3A27518E	ROM1:18, 119, 259-267; REG:242-266
Ethernet	MB	LAN8720A in REF_CLK-in mode, fed by the fan-out buffer. DNP 0 Ω and a DNP 25 MHz crystal for REF_CLK-out mode (strap nINTSEL=0, which shares the LED2 pin and flips its polarity). MDIO 1.5 kΩ pull-up. TXD0/1, TX_EN, MDC series resistors sit on the card. PHY reset from the host MCU. Record the PHY address straps. Human-routed 100 Ω MDI pairs to the MagJack	LAN8720A; MagJack; 1:2 fan-out buffer TBD	docs/bringup/KR260_LAN8720_FPGA_HUB_PLAN.md:4; HFW:74-94
REF_CLK	MB	50 MHz oscillator with OE (host MCU) into the fan-out buffer. Length-matched branches to the PHY and, through the slot, to RMTI_REF_CLK; count the card trace in the match. Weak pull-down on the slot side	TBD	SDC:13, 35-36
HOSTIO4 header	MB	7 signals plus ground; no power pin , because back-feeding the host's 3V3 is forbidden. Pin order HOSTIO[n] = RP2350 GPn, n = 0..6; this is the reference PIO driver's order, so confirm it against the bench build. 330 Ω series on the host side of the isolation links. IOACK pull-up about 5 kΩ and 10 kΩ idle pull-ups on the other six, on the slot side of the 330 Ω. Scope test points on IOACK and IOREQ1	Waveshare RP2350 (off-board)	PROC:163-246; hostio-pio.py:7-11
SWD header	MB	Arm 10-pin 1.27 mm Cortex debug layout. VTref = VDDIO; SWDIO and SWCLK with 22–33 Ω series. nRESET to the supervisor MR through a DNP link. Isolation links to the JTAG header	J-Link BASE (1.2–5 V)	XDBG:54-62

Block	Where	Requirement	Suggested part	Source
JTAG (boundary scan)	MB	TCK=SWDCK, TMS=SWDIO, TDI=HOSTIO[0], TDO=HOSTIO[1], TRST_n=SE, VTref, GND. Links make {HOSTIO header, JTAG} exclusive on HOSTIO[0:1] and {SWD, JTAG} exclusive on SWDCK/SWDIO (Table 1). Any J-Link-compatible connector that carries TRST	J-Link BASE; OpenOCD SVF	BSC:77-95; PTAB
Board-control MCU	MB	Reads CARD_ID and CARD_PRESENT_n before any rail, and waits for CARD_READY before NRST. Drives NRST (through MR), CLK_OE, REF_CLK_OE, MUX_SEL, EN_VDD, EN_VDDIO, EN_VIN_AUX and PHY_nRST. I ² C master for the INA228s, the TPS62869 and a programmable oscillator. Its I ² C bus must stay separate from the die's I2C_SCL/SDA. It can be the HOSTIO RP2350 or a second one on a control header	RP2350 (on a header)	–
TideLink park	Card	See Section 4.9. The 18 TL signals also reach the slot; the single-slot motherboard leaves them unconnected	–	BSC:455-476
Current monitors	MB	INA228 on VDD_DIE, VDDIO_DIE, VDDIO_BOARD and VIN_AUX (size that shunt for ≥ 4 A); optional on 3V3_AUX. Size each shunt so about 250 mA is full scale (start). Log energy and charge per phase	INA228	research report 03 §C3

4.5 Power

Rail	Nominal	Range on board	Die pads	Expected current	First-power limit (start)	Monitor
VDD_DIE (slot VDD_CORE)	1.20 V	1.00–1.25 V, 5 mV steps, Kelvin-sensed at the card. Never 1.32 V	6 × VDD, 4 × VSS (top and bottom)	≤ 50.7 mA (default-toggle estimate, 1.08 V SS, fp1505). Real workloads 7.4–8.3 mW at 50 MHz, 15.0 mW (about 14 mA) at 100 MHz. Leakage 0.36 mW (SS, 125 °C). 7.9–13.1 mA per core pad measured in rail analysis	100 mA	INA228
VDDIO_DIE	3.3 V	3.3 V; 2.5 V fallback	12 × VDDIO (1 POC), 12 × VSSIO	NOT FOUND. The IO ring was never analysed	50 mA	INA228
VDDIO_BOARD	= VDDIO	same regulator	–	Per the part datasheets (flash, PHY IO, oscillators, pull-ups)	–	INA228
VIN_AUX	5 V	fixed, eFuse	– (K26 card only; ≥ 6 contacts)	Up to about 4 A for the K26 card; design for ≥ 4 A continuous	eFuse	INA228
3V3_AUX	3.3 V	fixed	–	PHY analogue, board MCU if on-board	–	optional

Sources: docs/tapeout/31-power-delivery-measured.md:814-816,839-853,956; ASIC/eth-chiplet/build/power-saif-20260827/SUMMARY.txt:20-25; docs/tapeout/32-ir-drop-stage.md:802-810.

Sequencing. POC behaviour and power-sequencing rules are **NOT FOUND** (open question 2). The rule REG:473-477 gives for bring-up is: diode-check every pin, then measure IO supply current *before* applying core power. Every power-up therefore follows this standard sequence, S1, which the board-control script implements:

0. Card check (Section 3.5): both CARD_PRESENT_n low and CARD_ID = 000 (or 010). Otherwise stop: no rail is enabled. (The K26 card follows the same rule with its own rails: VDDIO and VIN_AUX, never VDD_CORE.)
1. CLK_OE = 0, REF_CLK_OE = 0, NRST held low (MR asserted), flash mux on side A, HOSTIO host outputs Hi-Z or host unplugged.
2. Enable VDDIO (current-limited). Wait for the INA228 reading to settle and record VDDIO_DIE current with VDD off. Keep this step short: what the POC does with the core unpowered is not known.
3. Enable VDD at 1.20 V (first time only: ramp in 0.1 V steps and record the current at each step). Record VDD_DIE current.
4. REF_CLK_OE = 1. Only now enable the HOSTIO host's pins or plug in the probe. The 330 Ω series resistors limit injection current if this order is broken.
5. Start the HOSTIO capture (HIO-001 needs the cold-boot banner).
6. Check CARD_READY (strapped ready on the die card; on the K26 card it follows configuration). Then release NRST with CLK **stopped** (erratum N1). Wait at least the supervisor timeout plus 10 ms.
7. CLK_OE = 1. Reset releases synchronously a few CLK edges later; the ROMs start.

Power-down reverses the order: CLK_OE = 0, NRST low, VDD off, VDDIO off. Only then may a card be removed or inserted.

4.6 Clocking

There is no PLL: the board oscillator is the fabric clock. The motherboard oscillator offers 25, 50 and 100 MHz (contract v1). Everything time-derived follows it: PHC NS_INCR, the QSPI boot clock (SCLK = CLK/2, because CLK_DIV resets to 1 and neither ROM writes it), the XiP no-flash watchdog ($65536/f$), timer reloads, and the firmware's FW_CORE_HZ.

CLK	NS_INCR = $10^9/f$	QSPI SCLK at boot	XiP watchdog	Use
25.0 MHz	40	12.5 MHz	2.62 ms	Option (contract v1): the rate every FPGA build runs at; K26 card in Phase M, and a slow point for the die Default and boot clock. Flash read +5.0 ns at SS with CLK_DIV = 1 (B2) Shmoo step Shmoo step About the slow-part limit (≈ 95 MHz at SS/cworst, item 6f). Flash read fails at CLK_DIV = 1 (-4.0 ns SS) SDC target. Flash read fails at CLK_DIV = 1 (-5.0 ns SS): firmware sets CLK_DIV ≥ 2 before CLK is raised
50.0 MHz	20	25 MHz	1.31 ms	
62.5 MHz	16	31.25 MHz	1.05 ms	
83.3 MHz	12	41.7 MHz	0.79 ms	
90.9 MHz	11	45.5 MHz	0.72 ms	
100.0 MHz	10	50 MHz	0.655 ms	

- RMII_REF_CLK is a 50 MHz *input*; the MII clocks are generated on the die at $\div 2$ SDC:13, 35-36.
- SWDCK is constrained to 25 MHz SDC:14. Start **boundary-scan TCK** at about 1 MHz BSC:134-138.
- The **TideLink link clock** is fixed at /1 and gated while ROLE_CFG is unwritten, which it always is on this board REG:151-153.
- The PHC NS_INCR_FRAC field is signed, and the procedure's FRAC formula is wrong when the fraction is 0.5 or more. That is the reason every frequency is $10^9/N$ NMS/ptp-hardware-clock-ahb/src/rtl/phc_clock_core.sv:132-156.

4.7 Reset

- NRST asserts asynchronously and releases synchronously on CLK into SYS_PORESETn/HRESETn, so **CLK must run for reset to release** PRMU:133-157.
- **Erratum N1.** Raw NRST also resets about 4,027 flops asynchronously, with no synchroniser: the PHC (899 flops) and the debug group. Releasing NRST with CLK stopped, as in sequence S1 step 6, is advised, not mandatory: with CLK running, STA puts 65-69 % of release phases at SS on a removal check, but every affected flop is benign docs/chip-reference-manual/review/closure/item06-sta.md §6a. Clear the perf counters and ETC and re-initialise the PHC after boot REG:359-377.
- The TideLink role lock clears only on power-on reset, i.e. NRST.
- Software resets live at 0x2A000020. Bit 0 is CPU0_SWRST. Bit 1 is CPU1_SWRST, which resets the whole fabric including HOSTIO, so the link drops and must be re-entered PROC:848; REG:280.

4.8 Debug headers and isolation links

Boundary scan and HOSTIO are mutually exclusive, and boundary scan shares its pins with SWD. Each bench session fits exactly one link set:

Session	HOSTIO[0:1] → HOSTIO hdr	HOSTIO[0:1] → JTAG hdr	SWD pins → JTAG hdr	SE	Notes
HOSTIO (P3–P7, P9–P15)	fitted	open	open	low	SWD header may stay connected (P8), subject to the two-master rule
SWD (P8)	fitted	open	open	low	SWD pins go to the SWD header
Boundary scan (P2)	open	fitted	fitted	high (last)	NRST low; HOSTIO host unplugged. Raise SE <i>before</i> driving TDI, or TDI fights a 16 mA driver BSC:20-40

Table 1. Isolation-link settings per session. “fitted” means the 0 Ω link or jumper is in place.

4.9 TideLink parking (on the die card)

All parking parts sit on the die card; the 18 TL signals and the I²C pair also reach the slot, so a later two-slot motherboard can join two dies, but the single-slot motherboard leaves them unconnected. The die cannot toggle these pins on its own. rc4 is built with NEG0_CFG_RESET0, SELF_ARM_TRAIN_EN1, AUTO_ANCHOR_EN1 and ROLE_FROM_STRAP1. Nothing on the die writes ROLE_CFG. While ROLE_CFG is unwritten, wlink_por_reset holds the link in reset and pad_clk_tx stays gated. The TX pads still drive a static level, but that level is not established REG:78-98.

- **TL_RX[7:0], TL_CLK_RX:** bond them, and pull each to VSSIO with 10 kΩ. They have no keeper.
- **TL_TX[7:0], TL_CLK_TX:** bond them (B-1) to test points and to the loopback footprints. They cannot be tri-stated and must never touch a rail.
- **Loopback footprints (B-1):** DNP 0 Ω TL_TX[n]→TL_RX[n] and TL_CLK_TX→TL_CLK_RX, next to the die and length-matched, keeping the 10 kΩ pull-downs. With them fitted, boundary-scan continuity of all nine lanes runs from the TAP alone (p2_extest_walk_tloop.svf), and after one ROLE_CFG write the die trains the shipped V2/CRC/ECC/8-lane configuration against itself through real pads at real CLK (P2b). This is the only silicon measurement of the D2D link possible before a compute die and a pair board exist docs/chip-reference-manual/review/closure/item03-d2d-loopback.md. In ring order the clock sits mid-bus: TL_CLK_TX is between TX[4] and TX[3], and TL_CLK_RX between RX[4] and RX[3] BSC:450-476.
- **I²C_SCL/SDA:** 4.7 kΩ to VDDIO on the card (the contract’s value; 2.2–4.7 kΩ is acceptable), plus test points. The pad comment says real 2.2 k pull-ups belong on the PCB PADS:926-932.
- **Firmware and tools:** never write ROLE_CFG 0x2E032080, except in P2b with the loopback fitted. Link status at 0x2E032084 is safe to read with the link down docs/design/PIN_MAP.md:196-210.

4.10 Die-card chip-on-board footprint and bonding rules

These rules apply to the **die card** only. The motherboard needs no bondable finish and can come from any fab.

Settle the bonder’s rules before the card layout. Zepler Cleanrooms (on campus) has a manual Au/Al wire bonder, dicing and die bonders. Its model and pad limits are unconfirmed; one search snippet describes a campus “5430 Fine Wire Bonder” as an ultrasonic wedge bonder (25 μm wire, 50 × 50 μm minimum pads), which would mean Al wedge and ENIG. Custom Interconnect Ltd (Andover) is the nearest outsourced house.

Rule	Value	Source
Wire and card finish	Al wedge (ultrasonic, unheated) works on ENIG : order the card as ENIG + gold edge fingers (Eurocircuits offers selective ENIG with a gold-edge-connector option). Au ball (thermosonic, heated substrate) needs ENEPIG ; ENEPIG plus hard-gold fingers on one card is unconfirmed (PCBWay lists both, not together). The bonder decides	Würth webinar 108; JLCPCB ENEPIG vs ENIG; research report 05 §4
Card fits the bonder	Check: the card must fit the bonder’s stage and clamp, and its heated stage if Au ball. An HB16-class manual bonder takes samples up to 100 × 150 mm [search snippet]; confirm Zepler’s limit and any thickness limit against the 1.60 mm card HSEC8 needs	research report 05 §4; Technion HB16 page
Bond finger	≥ 150 μm wide, 300 μm long; ≥ 100 μm between fingers (≥ 250 μm pitch)	Würth design-rule poster v1.2
Die edge to finger	≥ 1.5 × die thickness (thickness NOT FOUND)	Würth poster
Solder mask	One opening over the whole bond area	Würth poster
Wire length	25 μm Au ≤ 6 mm (17.5 μm ≤ 4.5 mm; 33 μm ≤ 7 mm)	Europractice packaging rules 2024

Rule	Value	Source
Wire angle	$\leq 45^\circ$ from the pad normal. No crossing wires	Europractice 2024
Die pad	Opening $\geq 55 \times 55 \mu\text{m}$ and pitch $\geq 65 \mu\text{m}$. This die has $58 \times 66 \mu\text{m}$ and $67/80 \mu\text{m}$ alternating	Europractice 2024; REG:500-502
Staggered rows	The P60 bond rules were never run (imec ran P80). The bonder must confirm its capillary and loop plan for $134/160 \mu\text{m}$ same-row pitch with a $102 \mu\text{m}$ row offset	REG:500-502; docs/tapeout/60-waiver-document-rc1v3r4.md:158-175
Die placement	$\pm 100 \mu\text{m}$ (manual)	Europractice 2024
Die attach	Conductive epoxy (H20E) to a VSS copper pad with the finish on it. Non-conductive epoxy (H70E) needs explicit ground bonds. Substrate and backside connection to confirm with imec	Europractice 2024
Glob top	Full, spot or dam-and-fill. Keep the keep-out ring free of parts and test points	Würth webinar
Fiducials	Not sourced; ask the bonder. Bonders usually photograph the die to find it	KiCad forum

Fan-out feasibility. Table 2 is a first check against these rules, computed by `figures/gen_figs.py`. Assumptions: one finger row at $250 \mu\text{m}$ pitch, centred on each die edge; distances measured from the seal-ring edge; each wire runs from the opening centre to the finger centre. The **left (D2D) edge governs**. A single rectangular finger ring at least 2.23 mm from the seal ring keeps every wire within 45° and under 3.4 mm long, inside the 6 mm limit. The left row's ends then overhang the die by 2.18 mm but stay clear of the top and bottom rows.

Edge	Pads	Die edge (mm)	Finger row (mm)	Overhang / side (mm)	Min. offset for 45° (mm)	Wire length at 2.23 mm (mm)
Left	26	2.04	6.40	2.18	2.23	2.3–3.3
Top	17	1.64	4.15	1.25	1.31	2.3–2.7
Right	22	2.04	5.40	1.68	1.73	2.3–3.0
Bottom	17	1.64	4.15	1.25	1.31	2.3–2.7

Table 2. Single-row fan-out estimate. The sawn edge lies beyond the seal ring by the scribe width, which is **NOT FOUND**; add it to every offset.

If the bonder wants shorter wires or a tighter angle, there are three options: two staggered finger rows; leaving `TL_TX[7:0]/TL_CLK_TX` unbonded (nine fewer left-edge wires, but no loopback and no P2b; rejected by B-1); or a finer finger pitch, if the bonder allows it.

Footprint generator. KiCad has no wire-bond object. Generate the footprint with a script from the pad CSV:

- Place the die pads at the **opening centres**, not the CSV centres. The CSV centre is the pad *cell* centre. Openings sit $36.8 \mu\text{m}$ (outer row) or $138.8 \mu\text{m}$ (inner row) from the design-frame edge; the seal-ring frame adds $20 \mu\text{m}$. Appendix C lists every opening.
- Correct the CSV edge label of `BuPAD_HOST_I0_6` from top to **right**.
- Emit: die outline and pads on a fab/documentation layer; fingers on F.Cu named by net; the VSS die-attach pad; the glob-top keep-out on a user layer; one solder-mask opening.
- Check: angle $\leq 45^\circ$, length $\leq 6 \text{ mm}$, no crossings, finger spacing $\geq 100 \mu\text{m}$, die-to-finger $\geq 1.5 \times$ thickness.
- Export a bonding diagram (pad, net, finger, wire length and angle) for Zepler.

Hedge. Package about 10 dies through Europractice in parallel with 2–3 COB die cards, and mount them on packaged-die cards (`CARD_ID 010`) in the same slot. Cost: €485 setup per bonding diagram, QFN Open-Pak €83–€229 per unit, so 10 in QFN $\approx$ €1.3k–2.8k. QFN needs dies $\leq 300 \mu\text{m}$ thick (Open-Pak $\leq 280 \mu\text{m}$), so thinning may be needed; ceramic takes up to $600 \mu\text{m}$. Lead time is not confirmed (up to 6 weeks in one search snippet). A bad COB die or bond scraps one small card; the motherboard is reused.

4.11 Test points

Test point(s)	Where	Net	Purpose
TP_VDD_T, TP_VDD_B	Card	VDD at the top and bottom fingers	Rail voltage at the die (Kelvin), for the shmoo
TP_VDDIO_x	Card	VDDIO, one per side	IO rail at the die
TP_GND $\times \geq 6$	both	VSS/VSSIO	Scope ground springs next to every signal test point group
TP_CLK	MB	CLK, slot side of the series R	Frequency; edges $\leq 1.2 \text{ ns}$; clean start after OE

Test point(s)	Where	Net	Purpose
TP_REFCLK	MB	RMII_REF_CLK, slot side	50 MHz present
TP_NRST	MB	NRST	Release timing against CLK (N1)
TP_SE, TP_TEST	Card	straps	Level check at the start of every session
TP_IOACK, TP_IOREQ1	MB	HOSTIO[2], HOSTIO[0]	Mandatory scope points: the only way to split “SoC never requested” from “host never acknowledged” PROC:188
TP_QSPI_x ×4	MB	QSPI_SCLK, nCS, IO0, IO1	Boot activity after NRST release; SCLK = CLK/2. B-5: also scope points at the die on SCLK, IO0, IO1 for the capture margin
TP_MDC, TP_MDIO	MB	RMII management	MDIO pull-up level; MDC rate
TP_TXEN, TP_CRSDV	MB	RMII	Frame activity
TP_TL_TX[7:0], TP_TL_CLK_TX	Card	TideLink outputs	Static level; EXTEST continuity. Unloaded otherwise
TP_SCL, TP_SDA	Card	die I ² C	Pull-up level
TP_CARD_ID, TP_PRESENT, TP_CARD_READY	MB	card detect	CARD_ID, mating and ready checks
TP_VIN_AUX	MB	VIN_AUX after the eFuse	K26 card supply
TP_CLK_OE, TP_MUX_SEL, TP_EN_x	MB	board control	Sequencing checks

4.12 PCB tooling

- **Flow:** SKiDL (Python; MIT; v2.3.0, 2026-07) or Diode Zener (pcb CLI) for the schematic and netlist → KiCad 10 → kicad-cli sch erc and kicad-cli pcb drc --format json --exit-code-violations in a loop that Claude fixes → Freerouting (CLI, v2.4.1) for the slow nets → KiCad/KiKit fab outputs. Two boards: the **motherboard** from any fab with a standard finish; the **die card** as ENIG + gold edge fingers (Eurocircuits) for Al wedge, or PCBWay ENEPIG ($\geq 2 \mu\text{in Au}$, wire-bond note on the order) for Au ball.
- **Claude does:** netlist, straps, BOM, the COB footprint generator, the HSEC8 pinout checker, the ERC/DRC loop, fab outputs.
- **A human does:** die and finger placement agreed with the bonder; the RMII, MDI and finger fan-out routing; magnetics placement; stack-up and impedance; final DRC sign-off.
- **Estimate** (research report 03): 1–2 days for the schematic, half a day for the footprint generator, 1–2 days for placement, 1–3 days for routing, then hours for DRC and fab outputs.

Tool comparisons, fab capabilities and MCP servers belong in a separate report.

#	Item	Check before power	Path	Status
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5 Preparation before first power

5.1 Bench equipment

- Two-channel bench supply with current limit, ESD-safe station, DMM with diode mode, 4-channel oscilloscope with bandwidth well above the 100 MHz CLK, stereo microscope and camera for bond photographs.
- The motherboard with its HSEC8 slot; the K26 FPGA card (Phase M, golden reference); the lab KR260s for emulator-bitstream development.
- J-Link BASE with OpenOCD. A Waveshare RP2350 running MicroPython with the HOSTIO PIO driver. A flash programmer for the off-board header. Optionally an SPI-NOR emulator.
- A host PC with a spare Ethernet NIC and Wireshark on a private subnet. apps/eth_ping_responder answers at 192.168.10.2.
- Optional: a thermal camera, and a temperature chamber for P15 (whether one exists is **NOT FOUND**).

5.2 Host tools checklist

#	Item	Check before power	Path	Status
1	RP2350 PIO driver	Status-nibble polarity fixed (the reference driver idles it active-low; the RTL expects active-high). put() guarded on tx_fifo() < 4. Resync watchdog live	NMS/nanosoc_arch_tech/rtl/hostio4/rpi-pico-pio/pico-mp/hostio-pio.py; PROC:247-280	exists
2	ADP client	First command after ESC is rejected (adp.enter() absorbs it). Parameters always x + 8 hex digits. Re-issue A before every access	RIG/hostio_adp.py	exists
3	HOSTIO suite	--target asic_tsmc65; capture to file with the die serial in the name. golden_asic_reference.json generated from the rc4-RTL dry run (build/presilicon-closure/item08/scripts/rig/eth_chiplet/hostio_suite/golden_asic_reference.json, sha256 98fcaa91...); place it in the repo	RIG/hostio_suite/docs/chip-reference-manual/review/closure/item08-bringup-dryrun.md	partial
4	Board-control script	Card check (CARD_ID, PRESENT) before any rail; wait for CARD_READY before NRST; sequence S1; INA228 logging; TPS62869 setpoint clamp ≤ 1.25 V; CLK_OE, NRST, MUX_SEL. Proven in Phase M	–	new
5	Address blocklist	Enforced in the client (refuses listed addresses); Appendix D	–	new
6	Flash programming flow	Programmer + mux control + read-back verify + sha256 logged	–	new
7	Boundary-scan script	IDCODE, BYPASS, SAMPLE/PRELOAD, EXTEST from the BSDL	sys_desc/bscan/nanosoc_eth_chiplet_pads.bsd1	new
8	SWD OpenOCD config	Manual AP setup (no ROM table); JTAG-to-SWD switch	FPGA reference docs/bringup/KR260_BOARD_WIRING.md:82	new

5.3 Flash images

Pack every image with flash_pack.py and validate it with check_flash.py, which checks the ROMs' own accept and reject rules. Log the sha256 of every image programmed.

Image	CPU1 (entry 1, slot A 0x30000)	CPU0 (entry 0, 0x60000)	Purpose / expected signature
IMG-0	– (erased device)	–	P4 boot autopsy with blank flash. Ladder rung 6 = 0x4
IMG-1	Minimal CPU1 image or the CPU1 shim. It must not write 0x2D000000 - 0x2D00004F or 0x2D001FF0	hostio_fw/fw_heartbeat built with TARGET=asic, FW_CORE_HZ = board CLK	P11. 0x2D001FF0 = 0xB007C0FE; 0x2D000004 = 0x5E27C0DE; 0x2D000008 = 0x48465702; 0x2D00000C = 0x02FAF080 at 50 MHz; 0x2D000000 advancing
IMG-2	as IMG-1	hostio_fw/fw_mdio_probe	P13.1–13.2. 0x2D000008 = 0x48465703; PHY ID at 0x2D000010/14; MODER and PACKETLEN at 0x2D000038/3C

Image	CPU1 (entry 1, slot A 0x30000)	CPU0 (entry 0, 0x60000)	Purpose / expected signature
IMG-3..	as IMG-1, or firmware partner	apps/* ASIC builds (P12-P14)	Results via <code>regr_sentinel.h</code> blocks in eth DMEM 0x18000000

Rules for every image:

- **TABLE0 only** for first silicon: no TABLE1 copy, no golden. `mkflash.sh` always adds TABLE1 (an identical copy) and a golden CPU1 image, so call `flash_pack.py` directly with the same offsets (`--app-offset 0x60000 --cpu1-slot A`) and without `--table1-offset/--golden` MKF:120-143.
 - Why TABLE0-only is safe: after two unconfirmed boots the CPU1 ROM demotes the primary table but still tries it last, so a lone TABLE0 keeps booting ROM1:344-362.
 - Why golden is left out: `flash_pack`'s golden table gives CPU0 a zero-size entry, which the CPU0 ROM rejects, so golden would recover CPU1 only `flash_pack.py`:326-329.
- TABLE0 must hold **2 or 3 entries**; the CPU0 ROM rejects any other count. Entry 0 is CPU0 (links at 0x0), entry 1 is CPU1 (links at 0x10000000) ROM0:128; MKF:43-57.
- **Flash-layout rule (FLASH-LAYOUT)**. No image may put anything in flash 0x00000-0x5FFFF except TABLE0, TABLE1, the counter sector (left erased) and CPU1 images in slot A/B or golden at their address-map offsets. XiP/overlay cold segments go at 0x80000: until 2026-10-07, 17 of 18 apps linked them at 0x20000, the boot-counter sector. Apply the NMS patch `build/presilicon-closure/item17/layout_fix_nms.patch` and check every image with `check_flash.py` with `--elf` on the app ELF `docs/chip-reference-manual/review/closure/item17-silicon-rom-envs.md`.
- Leave the counter sector at 0x20000 erased. Keep the whole image below 4 MB: the QSPI address register is 22 bits.
- Build firmware with `FW_CORE_HZ` equal to the real CLK. There is deliberately no default HFW:55-62.
- Before CPU0 boots, a CPU1 app must not write IPC slot 1 (0x23000010); if it does, CPU0's boot is only delayed, because its wait is bounded.

5.4 Record sheet

One sheet per board, filed with the capture files. Values marked * are one-shot: they can be taken only on the first power-up.

Field	Expected / reference	Recorded value
Motherboard serial; die-card serial; die ID / wafer position	-	
CARD_ID and CARD_PRESENT_n read before power	000; both low	
Bond date, operator, wire type, photo file	82 wires, no crossings	
P0 diode-check table (file)	Consistent with the board-only signature	
VDDI0_DIE current, VDD off	NOT FOUND (first data point)	
VDD_DIE current: CLK stopped, NRST low	Near leakage (0.36 mW at SS/125 °C)	
VDD_DIE current: running, 50 MHz	≤ 50.7 mA; real workloads 7-8 mW	
CLK at TP_CLK; f_HCLK (HIO-402)	Oscillator setting	
NRST release method	CLK stopped	
Banner word and arrival time*	0x50c1ab04, within 1 s	
Ladder rungs 0-6	P4	
Uninitialised SRAM read*	Open question 12	
Flash JEDEC ID at CLK_DIV 1, 2, 4	Fitted part	
Flash image and sha256	-	
Boot signatures	Section 5.3	
PHY ID; link; ping loss	0x0007/0xC0F1; up; 0 %	
Ambient temperature	-	

5.5 Address blocklist

The HOSTIO/ADP monitor has no HREADY timeout. One access to a stalling address wedges the die until a power cycle, and simulation can never reproduce it, so the blocklist is the only guard REG:198-208; PROC:1136-1147. The client and suite must refuse the entries in Appendix D unless an override names the test and a human is watching. The most important entries:

- 0x24000000 - 0x24FFFFFF, the QSPI XiP aperture (HZ-6).

- 0x2E0321A0 - 0x2E0321B8 (HZ-1).
- 0x2F000000 and up, the peer aperture (HZ-2).
- ROLE_CFG 0x2E032080: never written.
- Read-to-clear registers 0x2E032020/024/038/15C/160/168 (HZ-8).
- Spinlock acquire pages 0x2C100000/100/200 (HZ-7).

6 Phased bring-up procedure

Each phase ends in a **gate**; do not start the next phase until the gate passes. Phase M runs on the motherboard with the K26 FPGA card; P0 onwards run on the die card, in the same slot. **Path** codes: B = bench instruments; H = HOSTIO/ADP; J = JTAG; S = SWD; F = firmware result read back over HOSTIO; P = off-board programmer. **Status**: **exists** = tool or app exists; **partial** = exists but needs work; **new** = to be written; **proc.** = procedure only. HIO-*nnn* test IDs are in RIG/hostio_suite (T0 = tier0.py, T1 = tier1.py, T23 = tier2_3.py, T45 = tier4_5.py) and are specified in PROC.

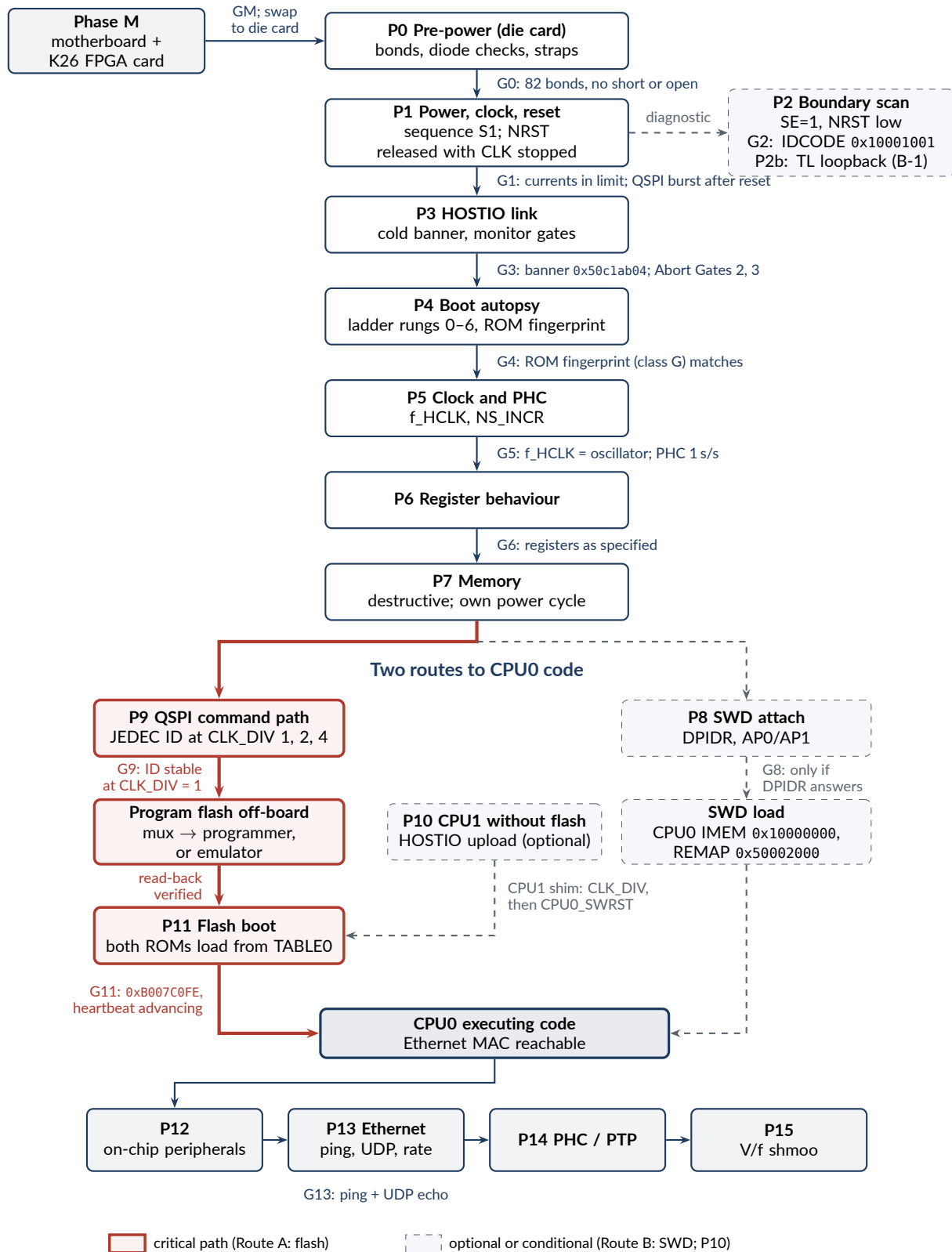


Figure 3. Phase flow and gates. Phase M proves the motherboard with the K26 FPGA card before the die card goes in. Route A (flash, red) is the plan of record. Route B (SWD) opens only if P8 reads a DPIDR. P10 can load a CPU1 shim that fixes CLK_DIV and re-boots CPU0, but CPU0 still loads its code from flash.

Phase M Motherboard bring-up with the K26 FPGA card

Purpose: prove the motherboard before any die is at risk. Every row below runs with the slot empty or with the **K26 FPGA card** (CARD_ID 001): a Kria K26I SOM on a slot-format carrier, running the eth-chiplet FPGA build on 3.3V LVCMOS IO at CLK = 25 MHz. The carrier is specified in FPGA_DAUGHTER_CARD_OPTIONS.pdf, in this directory. After Phase M the K26 card stays as the **golden reference**: repeat any later failure on it to separate a motherboard fault from a die fault.

Start now, before any board exists: develop the emulator bitstream on the lab KR260s (same XCK26 family). Undo the QSPI/HOSTIO4 tie-offs, take CLK from a pin, build with the silicon stage0_bootrom/stage0_bootrom_chip_core ROMs, and target the -i part (xck26-sfvc784-2LV-i). Start from the HAPS-SX board top, which already wires QSPI, HOSTIO, SWD and RMI and uses the silicon ROM. The KR260 + paddle adapter to the slot is **optional**: build it only if the motherboard is ready well before the K26 card.

#	Test	Pass criterion	Path	Tool / reference	Status
M.1	Slot empty: host MCU reads the card-detect pins	CARD_ID = 111, both CARD_PRESENT_n high; the script refuses every rail	B	board-control script	new
M.2	Rails at the slot pins (engineering override, slot empty or electronic load): VDD_CORE sweep 1.00–1.25 V into the DNP dummy load; VDDIO 3.3 V; VIN_AUX 5 V at 4 A	Each within tolerance at the slot; INA228 agrees with a DMM; a setpoint above 1.25 V is refused; current limits and the VIN_AUX eFuse trip as set	B	board-control script; INA228	new
M.3	Clocks at the slot: CLK at 25, 50 and 100 MHz; OE off and on; REF_CLK at the PHY and the slot	Frequency and edges (≤ 1.2 ns) correct; clean restart after OE, no runt pulse; CLK rests low while disabled	B	scope at TP_CLK, TP_REFCLK	proc.
M.4	Insert the K26 card (mechanical support fitted) with all rails off; power it	CARD_ID = 001, both PRESENT low. VDDIO and VIN_AUX on; VDD_CORE reads 0 V. Slot pull levels unchanged until CARD_READY is released (bus switches open: IOACK high, QSPI_IO1 low). The host waits for CARD_READY	B	board-control script	new
M.5	HOSTIO link through the header and the slot to the FPGA build	HIO-001 banner; Tier 0 passes against the FPGA target's reference values (25 MHz)	H	RP2350 + hostio_suite (FPGA target)	exists
M.5a	Card JTAG through the slot's spare pins	The motherboard's card-JTAG header sees the K26 chain	J	J-Link / Vivado HW server	new

#	Test	Pass criterion	Path	Tool / reference	Status
M.6	Scripted sequence S1 (NRST, CLK_OE, rail enables)	On the scope: NRST rises while CLK is stopped, CLK starts after the supervisor timeout; the FPGA build reacts if it takes NRST/CLK from the slot	B+H	board-control script	new
M.7	Flash path: mux to side B, program IMG-1, verify; mux to side A; read JEDEC ID and TABLE0 magic from the slot side	Programmer read-back identical. Through the slot: ID matches the part, magic 0x424F4F54	P+H	programmer flow; P9 script	new
M.8	PHY: link to the host PC; MDIO scan from the FPGA build	Link up at 100 Mb/s; PHY ID 0x0007/0xC0F1, stable (MDIO pull-up correct)	F+host	FPGA build firmware	partial
M.9	SWD and JTAG headers: continuity to the slot pins in each isolation-link set (rails off, card out)	Matches Table 1	B	DMM	proc.
M.10	Force one CARD_PRESENT_n high at TP_PRESENT while powered (do not pull the card)	The host drops every slot rail at once	B	board-control script	new

Gate GM. Rails, clamp, eFuse and card-detect/CARD_READY gating, all three clocks and OE, the HOSTIO link, card JTAG, the flash path through the slot, the PHY link and the scripted S1 all work on the K26 card. Then power everything down, remove the K26 card, and insert the die card for P0.

P0 Pre-power (die card bonded, not powered)

Purpose: find shorts, opens and wrong straps before any current flows. Rows 0.2–0.6 run on the die card *out of the slot*, measured at its edge fingers, so only the card's own parts load the readings. ERC and PadShorts were never run on rc4, and LVS is blind at the pad boundary REG:464-477, 497-499.

#	Test	Pass criterion	Path	Tool / reference	Status
0.1	Host stack dry run: status-nibble fix, resync watchdog, capture to file	Verified in Phase M against the FPGA card	B	PROC:247-280	exists
0.2	Bare-card signature <i>before bonding</i> : DMM resistance and diode reading from every card-edge finger	Matches the card schematic (pull values; no finger-to-finger short)	B	this plan	proc.
0.3	Bond inspection and photographs	82 wires, no crossings, no wire near the die edge	B	bonder	proc.
0.4	Diode check of every signal finger to VSSIO and to VDDIO, compared with 0.2	Every signal pad adds the die's ESD-diode signature; no short, no missing diode (an open bond); consistent across boards	B	REG:473-477	proc.
0.5	Rail-to-rail resistance at the card fingers: VDD-VSS, VDDIO-VSSIO, VDD-VDDIO	No short; values recorded	B	–	proc.
0.6	Strap and population check	Card: TEST low, SE pull-down, TL_RX pulls fitted, loopback links not fitted, CARD_ID straps 000. Motherboard: SE jumper off, mux on side A, links in the HOSTIO set (Table 1)	B	Section 3.7	proc.
0.7	Insert the die card with every rail off	Both CARD_PRESENT_n low; CARD_ID reads 000	B	board-control script	new

Gate G0. All rows pass. On any short or open: stop, photograph, and re-inspect the bonds before applying power. Apply the glob top only after G0 passes (a bond can still be lifted to isolate a pad before then).

P1 Power, clock and reset

Purpose: bring up the rails with current limits and start the die with the N1-safe sequence S1 (Section 4.5).

#	Test	Pass criterion	Path	Tool / reference	Status
1.0	S1 step 0: card check	Rails are enabled only after CARD_ID = 000 and both PRESENT low	B	board-control script	new
1.1	VDDIO only (VDD off), limit 50 mA	Stable; no current limit hit. Record it (no reference value exists)	B	INA228	proc.
1.2	VDD ramp to 1.20 V in 0.1 V steps, CLK stopped, NRST low, limit 100 mA; read the voltage at the card (Kelvin sense)	Monotonic, near leakage. Stop above about 10 mA static (start value)	B	INA228; REG:403-405	proc.
1.3	Thermal check	No hot spot on the die or the regulators	B	thermal camera	proc.
1.4	CLK on (50 MHz) with NRST still low	TP_CLK frequency and swing correct, edges ≤ 1.2 ns; record VDD_DIE in reset	B	scope; SDC:66-67	proc.
1.5	S1 steps 6–7: stop CLK, release NRST, start CLK	VDD_DIE current rises and settles below 50.7 mA. A QSPI_nCS and SCLK burst at CLK/2 appears within milliseconds: CPU1's ROM is running	B	scope on TP_QSPI_x; REG:170-176, 227-252	proc.

Gate G1. Currents are within limits, CLK is clean, and a QSPI burst follows reset release. If there is no QSPI activity, CPU1's ROM is not running: check CLK, NRST, SE and TEST with the scope before touching HOSTIO.

P2 Boundary scan (own session; diagnostic)

Session setup: NRST low; HOSTIO host unplugged; boundary-scan links fitted (Table 1); SE raised *before* TDI is driven; TCK about 1 MHz. Every net on HOSTIO4_P1[6:2], QSPI_IO[3:0] and RMII_MDIO high-Z on the board side (the walk drives them). Ready-made SVFs, generated from the BSDL and replayed clean against the RTL; the rc4 netlist passes the gate bench 7/7: p2_idcode.svf, p2_bypass.svf, p2_sample.svf, p2_extest_walk.svf, p2_extest_walk_tlloop.svf (in build/presilicon-closure/item14/svf/; docs/chip-reference-manual/review/closure/item14-bscan.md). PIN_MAP in the BSDL still needs the card nets (B-7). Boundary scan works with the core dead, so run it early if P3 fails.

#	Test	Pass criterion	Path	Tool / reference	Status
2.1	IDCODE, 64-bit shift	0x10001001, IR length 4	J	BSC:201-231	new script
2.2	BYPASS (1111); boundary-register length	Exactly 1 stage; 76 cells	J	BSC:254-271	new
2.3	SAMPLE, then PRELOAD and EXTEST: drive pattern P and its inverse, then walk a 1 through every driven cell	Outputs toggle as loaded, seen at the test points. Input cells capture board levels; they are observe-only	J	BSDL sys_desc/bscan/*.bsdl; BSC:332-412	new
2.4	With the B-1 loopback fitted: p2_extest_walk_tlloop.svf	All nine lanes continuous, clock mid-bus in ring order	J	BSC:450-476	new

Gate G2 (diagnostic). IDCODE reads correctly: the pad ring and the TAP are alive. A failure here does not block P3, but record it. At session end: SE low, links back to the HOSTIO set. HIGHZ is not implemented and CLAMP is untested.

P2b TL loopback (new, B-1; loopback footprints fitted; diagnostic)

Runs after P3 has a working HOSTIO link and P2's 2.4 has shown the nine lanes continuous. One die is both ends of its own link: its TL_CLK_TX is its RX clock. Bring it up at the lowest link clock (CLK = 50 MHz) and scope TL_CLK_RX at the die first: one lost link-clock edge is enough to wedge the link docs/chip-reference-manual/review/closure/item07-d2d-traffic.md.

#	Test	Pass criterion	Path	Tool / reference	Status
2b.1	Fit the 0 Ω links; power cycle; S1	P1–P3 results unchanged	B	–	proc. new
2b.2	Write ROLE_CFG 0x2E032080 = 0x02 (die_a) or 0x03 (die_b) over HOSTIO; poll link status 0x2E032084	cal_done and all 7 FC nodes at state 4; in rc4-RTL sim at 23.7 ms at 50 MHz, with no further host steps	H	docs/chip-reference-manual/review/closure/item03-d2d-loopback.md	new
2b.3	Program the CAM (0x2F→0x2D); write 16 words through 0x2F001000 (word, halfword, byte), read back through 0x2D and 0x2F with the M0+ or HOSTIO, never the DMA-250	Byte-exact, no bus error, CRC error counters 0	H/F	docs/chip-reference-manual/review/closure/item03-d2d-loopback.md	new
2b.4	If tl_data_mode stalls or a lane shows a persistent sync error just after training: re-arm training and count retries	At most one retrain (CDC class D: calibrator→TX single-flop; R1 reset release)	H	docs/chip-reference-manual/review/closure/item12-cdc.md §4	new

Gate G2b (diagnostic). The die trains against itself and moves data byte-exact: the shipped D2D configuration works through real pads at real CLK. A peer bus error or timeout means POR (power cycle) and retry, never a one-sided retry. A failure does not block P3–P15; remove the links afterwards if they load the TX pads.

P3 HOSTIO link

#	Test	Pass criterion	Path	Tool / reference	Status
3.1	HIO-001 cold banner (capture ≥ 10 s from S1 step 5)	Exactly one 0x50c1ab04 within 1 s. 0x50c1ab01 = wrong (ADPBASIC) build. Several banners = die resetting	H	PROC:298-320; T0	exists
3.2	HIO-002/003/004/010: monitor entry, bad-command rejection, exit and re-entry	Abort Gate 2 passes	H	PROC:340-346; T0	exists
3.3	HIO-005/006/008/009/011/120: address register, size encoder, auto-increment, block dump	All pass. If HIO-006 fails, every later access has the wrong size: stop	H	T0	exists
3.4	HIO-007 R! control pair	0x30000000 returns R!; 0x08000000 returns 0x18003c00	H	PROC:346; T0	exists

Gate G3. Banner plus Abort Gates 2 and 3. **No banner (Abort Gate 1)** does *not* mean a dead die. Six of the fourteen causes in PROC §1.6 give this symptom: wrong pin map, floating CLK or NRST, SE high, the wrong host driver polarity, among others. Scope TP_I0ACK and TP_I0REQ1 and work PROC §9.1 before concluding anything about silicon.

P4 Boot autopsy and identity

Read the ladder before anything else and record every rung. With blank flash (IMG-0) and the QSPI_IO1 pull-down, expect all rungs present and rung 6 = 0x4. The HardFault diagnosis in PROC §0.3 is superseded by REG: 227-253: a missing flash does not fault CPU1.

#	Test	Pass criterion	Path	Tool / reference	Status
4.1	HIO-106 ladder: rung 0 0x800000E4; 1 0x98000200; 2 0x21000030; 3 0x21000000[8]; 4 0x23000010; 5 0x98000084; 6 0x29000000	0x08000678; 0x0F1C0DE1; 0x0000800B; 1; 0xD15C0001; 0xFFFFFFFF (blank) or table data; 0x4 (no image) or 0x5 (CPU1 app). If rung 6 reads 0, re-read after 60 s	H	PROC:86-97; T1	exists
4.2	HIO-101/102 ROM fingerprint (class G, mask-verified)	e.g. eth ROM +0x0E4 = 0x0800048c; cc ROM +0x0E4 = 0x08000678 (full list PROC §8.4)	H	PROC:995-1015; T1	exists
4.3	HIO-103-127 identity; HIO-114 census	As PROC; eth DMEM word 0 reads 0x00000000 on the ASIC (not the FPGA value)	H	PROC:125; T1	exists
4.4	One-shot*: read uninitialised SRAM on the first power-up	Record; decides whether HIO-107's replacement works	H	PROC:1172	proc.

Gate G4. The ROM fingerprint matches; a class-G mismatch is a hard finding. The ladder is classified. If rung 4 is published and rung 6 still reads 0 on a re-read, that is the only silicon finding in the ladder.

P5 Clock and PHC

#	Test	Pass criterion	Path	Tool / reference	Status
5.1	HIO-402: f_HCLK from the perf probe over a host-timed interval ≥ 15 s	Matches the oscillator within its tolerance	H	PROC:447-467; T45	exists
5.2	Write NS_INCR = $10^9 / f$ to 0x22000008 (20 at 50 MHz); HIO-403/404	PHC advances 10^9 ns per second within oscillator tolerance. Reset value 4 runs time at $0.2 \times$ at 50 MHz	H	PROC:595-619; REG:341; T45	exists
5.3	HIO-311/406: CPU1 timers	Count down	H	T23/T45	exists

Gate G5. The measured clock equals the oscillator, and PHC time runs at 1 s/s. Derive every later timing figure from the measured f, not the nominal one.

P6 Register behaviour

#	Test	Pass criterion	Path	Tool / reference	Status
6.1	HIO-301–314. Gate 8 is HIO-313, with its control at 0x2100000C	As specified	H	PROC:536-585; T23	exists
6.2	HIO-305 mailbox; HIO-304 spinlock (DBG page 0x2C100200 only)	As specified; no lock left held	H	PROC:591; T23	exists
6.3	HIO-411 DMA-250 ID at 0x20000FC8	0x2500043B: present and not power-gated	H	PROC:1034; T45	exists

Gate G6. All pass. A B-class row that goes red is a documentation finding until the RTL proves otherwise PROC:1126.

P7 Memory (destructive; separate power cycle)

Tier-2 tests wipe eth IMEM and shared SRAM. Run them in their own power cycle, and never before a firmware-dependent test in the same power cycle PROC:1147.

#	Test	Pass criterion	Path	Tool / reference	Status
7.1	HIO-118, HIO-201–211 on all five RAMs: eth IMEM 32 KB, eth DMEM 16 KB, CPU1 IMEM 16 KB, CPU1 DMEM 8 KB, shared SRAM 8 KB	Walking 1s and 0s, address-in-address and fill all pass. About 1.5 h with full dumps	H	PLAN:496-506; T23	exists
7.2	At-speed March test in firmware	No failures (there is no MBIST)	F	–	new

Gate G7. No memory faults. Power-cycle before P8 onwards.

P8 SWD attach (Route B; not on the critical path)

#	Test	Pass criterion	Path	Tool / reference	Status
8.1	J-Link BASE, JTAG-to-SWD switch sequence, read DPIDR. Keep SE = 0. Leave HOSTIO idle (two-master rule)	A DPIDR is read. The expected value is NOT FOUND : record what comes back	S	NMS/docs/SWD_REFLASH_ASIC.md; REG:215-216	new config
8.2	AP0 (CPU0, 0xA0000000) and AP1 (CPU1, 0xB0000000) by hand (no ROM table); single accesses with read-back	Halt and resume both cores; read the MAC MODER at 0x40000000	S	REG:313-319, 342	new
8.3	If 8.1 passes: halt CPU0, load an image at 0x10000000, set REMAP 0x50002000 = 1, resume	CPU0 image signature appears	S	NMS/docs/SWD_REFLASH_ASIC.md:289-328	partial

Gate G8. A DPIDR is read: Route B is open. Otherwise record the result and continue on Route A. SWD failing is not a blocker.

P9 QSPI command path

Run at **CLK = 50 MHz**: STA predicts +5.0 ns read margin at SS with CLK_DIV = 1, and a fail at 100 MHz [docs/chip-reference-manual/review/closure/item06-sta.md](#) §6c. Use the APB command path only; **never touch 0x24xxxxxx** (HZ-6). Program the flash off-board through the mux, never from HOSTIO on the first die [PROC:1143](#).

#	Test	Pass criterion	Path	Tool / reference	Status
9.1	Read the JEDEC ID through 0x21000008/0x21000010 at CLK_DIV = 1, 2, 4	ID matches the fitted part at every divider. The two in-tree FPGA notes on the command format disagree	H	REG:276-279; HIO-312/115	new script
9.2	After off-board programming, read the TABLE0 magic at flash 0x000000 through the same path	0x424F4F54 ("BOOT")	H	AMAP:393	new

Gate G9. Reads are consistent at CLK_DIV = 1, the ROM's rate (SCLK = CLK/2). If only CLK_DIV ≥ 2 works at 50 MHz, the ROM boot will fail and the fault is the board or flash, not the predicted die limit. Lower CLK, or use REG §3.1 workaround 2 (write CLK_DIV, then pulse CPU0_SWRST) or 3 (the CPU1 shim).

P10 CPU1 without flash (optional; sim-proven, never run on a die)

Two hazards: setting 0x29000000 bit 0 is irreversible until power-on (HZ-4), and CPU1_SWRST drops the HOSTIO link (HZ-5). The docs disagreed on whether this route exists ([REG:258](#) and [PROC:850-873](#) say yes; [HFW:319](#) says no). The rc4-RTL dry run settles it: **the route works** (CPU1 runs the uploaded image) [docs/chip-reference-manual/review/closure/item08-bringup-dryrun.md](#). But the repo's fw_cpu1_park.bin writes no signature, so 10.2 cannot pass until fw_cpu1_park.S writes 0x48465701 to 0x2D000008.

#	Test	Pass criterion	Path	Tool / reference	Status
10.1	Four-command write/read-back at 0x90003ff0	Value sticks	H	PROC:866-873	exists
10.2	Upload image to 0x90000000; set 0x29000000 bit 0; 0x2A000020 = 2; re-enter the link	Park-image signature 0x48465701 at 0x2D000008	H	PROC:683-742; the suite refuses HIO-508 tier4_5.py:850	new shim

Gate G10. CPU1 runs uploaded code. That opens the CPU1-shim route: set CLK_DIV, AHB_SETUP and XIP_ACTIVE, publish 0xD15C0001, then pulse CPU0_SWRST. CPU0 still needs a valid flash image.

P11 Flash boot (critical path)

Run at **CLK = 50 MHz**. Raise CLK only after P11 passes and firmware has set CLK_DIV ≥ 2: at CLK_DIV = 1 the read fails above ≈66 MHz at SS. Check the image with check_flash.py before programming (flash-layout rule, Section 5.3).

#	Test	Pass criterion	Path	Tool / reference	Status
11.1	Program IMG-1 off-board (mux on side B), verify by read-back, set the mux to side A	Read-back identical; sha256 logged	P	–	new flow
11.2	Power-cycle with S1; read the ladder	Rung 6 = 0x5: CPU1 is running its app	H	PROC:86-97	exists
11.3	Read the CPU0 signatures	0x2D001FF0 = 0xB007C0FE; 0x2D000008 = 0x5E27C0DE; 0x2D000008 = 0x48465702; 0x2D00000C = FW_CORE_HZ; 0x2D000000 advancing	H	HFW:74-94, 426-436	exists
11.4	Five further power cycles	Boots 5/5. The primary table is retrieved last after the counter trips	H	ROM1:344-362	proc.

Gate G11. CPU0 runs code from flash, so the MAC is now reachable: the critical path is closed. If P9 passed but P11 fails, run check_flash.py on the programmed bytes, then check entry order and the FW_CORE_HZ build.

P12 On-chip peripherals via firmware

Results go to DMEM through `regr_sentinel.h`, read over HOSTIO at `0x18000000`. Apps live in `NMS/firmware/apps/`; each needs an ASIC build packed into a flash image.

#	Test	Pass criterion	Path	Tool / reference	Status
12.1	NVIC, peripheral IRQs, CPU0 timer and watchdog	Sentinel pass	F	<code>network_core_nvic_selftest</code> , <code>network_core_periph_irq_delivery</code>	partial
12.2	CPU1 watchdog reset and RESET_INFO	Reset cause recorded (HIO-302)	F+H	<code>chip_core_wdog_reset</code>	partial
12.3	IPC doorbell both ways; spinlock hammer	Doorbell fires once; no lost updates	F	<code>ipc_irq_*</code> , <code>ring_2core_hammer_*</code>	partial
12.4	DMA-250 copy within one port only , verified by read-back	Data correct; DONE alone is not proof (HSEL erratum)	F/H	<code>dma250_done_semantics</code>	partial
12.5	CRC accelerator	Matches the software CRC	F	<code>crc_accel_selftest</code>	partial

Gate G12. Every row passes or is logged against a known erratum (Section 8).

P13 Ethernet

Needs the 50 MHz REF_CLK and the PHY. Rebuild the apps for the board CLK: the MDIO divider is hard-coded for 25 MHz.

#	Test	Pass criterion	Path	Tool / reference	Status
13.1	MAC reset values (IMG-2)	<code>MODER = 0x0000A000</code> , <code>PACKETLEN = 0x00400600</code>	F	<code>hostio_fw/fw_mdio_probe.c</code> ; PROC:879	exists
13.2	MDIO scan	Stable PHY ID <code>0x0007/0xC0F1</code> at the strapped address; zero BUSY time-outs	F	same; HFW:78-93	exists
13.3	MAC internal loopback	Frame returns byte-exact	F	<code>eth_mac_loopback</code>	partial
13.4	PHY loopback; link and autonegotiation	Frames seen; link up at 100 Mb/s	F	<code>eth_tsu_watermark</code> , <code>eth_phy_link</code>	partial
13.5	ARP and ICMP ping from the host	Replies; 0 % loss over 1000 pings	F+host	<code>eth_ping_responder</code> (192.168.10.2)	partial
13.6	UDP echo, port 7	Echo returns byte-exact	F+host	<code>eth_netapp</code> (PicoTCP)	partial
13.7	Throughput	Rate measured and recorded	host	Scapy scenarios exist	new

Gate G13. Ping and UDP echo work: this is the board's main deliverable. Do not toggle LOOPBCK or TXEN mid-traffic (no CDC sign-off, REG N5); poll rather than rely on the TX-complete IRQ.

P14 PHC and PTP

#	Test	Pass criterion	Path	Tool / reference	Status
14.1	PHC register and alarm coverage; signed FRAC trim	Pass; rate within the ppm target	F/H	<code>phc_selftest</code> , <code>phc_reg_coverage</code> , <code>phc_alarm_irq_pulse</code>	partial
14.2	PTP slave against a host grandmaster	Offset converges	F+host	<code>ptp_slave</code> : hard-codes <code>NS_INCR = 40</code> and reports over a UART with no pad, so port it first	partial

PPS output cannot be tested: it has no pad.

P15 Characterisation

See Section 9.

7 Firmware, applications and host tools to build

Efforts are engineer-days, from research report 02 unless marked ^P (this plan's estimate). "Needed by" is the first phase that cannot run without the item.

#	Item	Status	Path	Effort (d)	Needed by
Tier A: before first power					
0	K26 FPGA card (slot carrier) and the emulator bitstream for the -i part, developed on the lab KR260s now	separate study	FPGA_DAUGHTER_CARD_OPTIONS.pdf (this directory)	see study	Phase M
1	HOSTIO host adapter re-verify (RP2350 PIO driver)	exists	NMS/nanosoc_arch_tech/rtl/hostio4/rpi-pico-pio/pico-mp/hostio-pio.py	0.5	P0
2	HOSTIO client and suite, ASIC target; place golden_asic_reference.json (generated by closure item 8)	partial	RIG/hostio_adp.py, RIG/hostio_suite	1	P3
3	Board-control library: CARD_ID/PRESENT/READY gating, VIN_AUX eFuse, sequence S1, INA228 logging, TPS62869 clamp, CLK_OE, NRST, mux, PHY reset	new	–	1–2 ^P	Phase M
4	Phase sequencer and record sheet (minimal first, full by P11)	partial	suite harness; PROC §8	2–3	P1
5	Address blocklist enforced in the client	new	–	(in 2)	P3
6	Boundary-scan script	new	BSD_L sys_desc/bscan/nanosoc_eth_chiplet_pads.bsdl	1	P2
Tier B: critical path to CPU0 code					
7	ASIC flash-image recipe (TABLE0-only) with CPU0 fw_heartbeat/fw_mdio_probe built for the board CLK	exists (tools)	flash_pack.py, MKF, check_flash.py, RIG/hostio_fw	0.5	P11
8	Off-board flash programming flow (programmer, mux, verify)	new	–	0.5–1 ^P	P9
9	CPU1 image with a signature outside the hostio_fw ABI range	new	fixes the 0x2D000000 clash with chip_core_flashmark	(in 7)	P11
10	CPU1 park image with its signature word	partial	RIG/hostio_fw/fw_cpu1_park.S	0.1	P10
11	CPU1 bring-up shim: CLK_DIV, NS_INCR, 0xD15C0001, CPU0 release, signature, counter	new (specified)	REG:286-295	1	P10 / G9 fallback
Tier C: peripherals and Ethernet					
12	Console replacement: shared-SRAM log ring plus regr_sentinel.h result blocks	new	NMS/firmware/include/regr_sentinel.h	1	P12
13	At-speed March memory test	new	–	1	P7.2
14	SWD OpenOCD config for the ASIC (no ROM table)	new	FPGA ref.	0.5	P8
15	Ethernet apps: DMIC builds, MDIO divider, PHY, DMEM telemetry	partial	docs/bringup/KR260_BOARD_WIRING.md:82 NMS/firmware/apps/eth_*	2	P13
16	Throughput tool	new	Scapy scenarios NMS/ethernet-subsystem-ahb/firmware/eth_app/tests/scenarios	1	P13.7
17	PTP demo: port ptp_slave (NS_INCR, UART → DMEM); host grandmaster set-up (NOT FOUND)	partial	NMS/firmware/apps/ptp_slave	2–3	P14
Tier D: characterisation					
18	Shmoo harness (TPS62869 + oscillator control, re-run set, pass/fail map)	new	–	3–5 + instruments	P15
Not needed for the first die					
19	OpenOCD over HOSTIO (memory access only)	exists	RIG/openocd_hostio4/hostio4.c	0	optional
20	HOSTIO flash programmer (qspi_flasher ported to CPU1 plus a host driver)	partial	NMS/firmware/apps/qspi_flasher	2–3	later dies

Total. Items 1–18 (the FPGA card, item 0, is costed in its own study) come to about **18–24 engineer-days** (4–5

engineer-weeks). Tier A is 5.5–7.5 days and must finish before the first die is powered. Tier B is 2–2.6 days and is on the critical path to Ethernet. Tiers C and D can overlap with bench time. PCB-side scripts (schematic, footprint generator, ERC/DRC loop) are counted separately in the PCB-tooling section.

Ordering. Tier A in parallel with PCB layout. Then Tier B, starting with items 7 and 8 so that IMG-1 is ready when the boards return. Then items 12, 13 and 15, which unlock P12 and P13. Items 14, 16 and 17 as bench time allows. Item 18 last.

8 Errata and workarounds

IDs are those of the rc4 issue register (REG) where one exists. “Board” marks an item this board’s design addresses directly.

ID	Issue	Effect here	Workaround / board rule	Source
B2	QSPI CLK_DIV resets to 1 and neither ROM writes it	Boot reads flash at SCLK = CLK/2. Corrected STA (10-08): data is captured on the rising SCLK edge; reads fail at CLK 100 and 90.9 MHz on every corner, pass at 50 MHz; safe $\leq \approx 66$ MHz docs/chip-reference-manual/review/closure/item06-sta.md	Board: boot at 50 MHz CLK; raise CLK only after CLK_DIV ≥ 2 . Or write CLK_DIV then pulse CPU0_SWRST (HOSTIO), or use the CPU1 shim	REG: 170-196, 268-298
–	Blank or absent flash: busy-bit poll up to 2^{24} times	Boot takes 27.5 s at 50 MHz if IO1 floats high (1.0ms with the pull-down; rc4-RTL sim); 0x29000000 reads 0 meanwhile. PROC \$0.3’s HardFault prediction is wrong	Board: QSPI_IO1 pull-down. Re-read after 60 s before classifying a 0	REG: 227-253
B3	ADP has no HREADY timeout; HOSTIO, NRST and CLK have no keepers	A stalling address wedges the die until a power cycle; a floating-low IOACK sticks the FSM	Board: IOACK pull-up; bias on all seven HOSTIO lines. Blocklist; read back the last beat of every F/U	REG: 198-208; PLAN: 318-323
B4	Firmware only from flash; no UART; SWD unproven on a die (answered on FPGA)	Nothing reaches the MAC without provisioned flash	Board: flash mux, programmer header, emulator header	REG: 210-217, 254-266
B1	No resident ROLE_CFG writer	Standalone: wanted. The link stays in reset and TX clock gated	Never write 0x2E032080, except in P2b	REG: 76-98
D2D-DMA	DMA-250 must never target 0x2E000000 - 0x2FFFFFFF (new, 2026-10-05)	Peer→peer: writes land twice, reads return 0; local→peer: each word written 2–3 times; peer→local: wrong data. DONE with no error	Never point the DMA-250 at the D2D window; move peer data with M0+ LDR/STR. Here: no peer die, so P12.4 stays within one port	docs/chip-reference-manual/review/closure/item11-tl053.md ; docs/tapeout/67-rc4-issue-register.md
RX-POL	TideLink RX polarity out_prepend_swi_polarity must stay at its reset value 1 (new, 2026-10-09)	Falling-edge RX capture accepts compute link 0’s TX with ≥ 2.70 ns; rising-edge capture fails hold by up to 0.85 ns at any clock rate	Firmware never clears it (P2b, and every pair phase)	docs/chip-reference-manual/review/closure/item06g-d2d-rx-window.md
FLASH-LAYOUT	XiP/overlay apps linked cold code at flash 0x20000, CPU1’s boot-counter sector (firmware defect, not silicon)	Each boot the ROM may program a byte of that code to 0x00, or demote TABLE0 at two leading 0x00; latent in current builds	Cold segment at 0x80000 (nanosoc_xip_layout.ld, patch pending); flash-layout rule; check_flash.py with - -elf	docs/chip-reference-manual/review/closure/item17-silicon-rom-envs.md
R1-R3	D2D peer port has no recovery; one dropped link-clock edge wedges the link with every FCSM still at 4	Backstop returns one ERROR after ≈ 1.55 ms; nothing clears it	POR both dies (here: the one die) and re-train on any peer error or timeout; detect a wedge with a canary read	docs/chip-reference-manual/review/closure/item07-d2d-traffic.md
HSEL	Two subsystem ports skip HSEL qualification	A cross-port DMA-250 copy silently corrupts CPU1 memory or repeats 0x2xxx writes while still reporting DONE; concurrent debug accesses can be dropped	DMA within one port only; verify by read-back; no SWD or HOSTIO access to that subsystem during DMA. Never two of {DMA, DAP, ADP} on one subsystem at once	origin/docs/errata-cpu-ss-hsel: TAPEOUT_LIMITATION_CPU_SS_HSEL.md (not on this branch)

ID	Issue	Effect here	Workaround / board rule	Source
N1	Raw NRST async-resets about 4,027 flops (PHC, debug group) with no synchroniser	Skewed or metastable reset if NRST releases near a CLK edge; rc4 SS-SDF GLS puts the window at about ± 1 ns of a CLK rising edge docs/chip-reference-manual/review/closure/item10-gls.md	Board: release NRST with CLK stopped (sequence S1; advised, B-4: STA finds every affected flop benign); re-initialise the PHC after boot	REG:359-377; docs/chip-reference-manual/review/closure/item06-sta.md
-	PHC NS_INCR resets to 4	Time runs at $4f/10^9$ with no error ($0.2\times$ at 50 MHz)	Write $10^9/f$ to 0x22000008	REG:341
-	PHC NS_INCR_FRAC is signed	The procedure's FRAC formula is wrong for fractions ≥ 0.5	Board: choose $f = 10^9/N$	phc_clock_core.sv : 132-156
-	Debug AHB bridge always returns OKAY; 15/16 of each 16 MB window aliases; back-to-back accesses drop	Stale data looks valid	Single accesses with read-back	REG:313-319
-	CoreSight ROM table absent	The debugger cannot discover the part	Configure APs by hand	REG:342
S1	Debug permanently unlocked	Security only; deliberate	None; docs/design/PIN_MAP.md :188 is wrong	REG:300-311
T1	No scan, no MBIST; ICG TE tied low	Functional test only	At-speed March test (P7.2)	REG:321-326
-	Boundary scan: HIGHZ not implemented; 15 outputs cannot be tri-stated; inputs observe-only (no INTEST); CLAMP untested	EXTEST continuity only	Board: test points on outputs; DNP loopback for TL lanes	REG:328-333; BSC:400-412
N5	No CDC sign-off on the shipped RTL (MAC IRQ, PTP queue)	Intermittent IRQ or timestamp faults	Poll; do not toggle LOOPBACK/TXEN mid-traffic	REG:419-431
F1	ERC never executed; PadShorts never run on rc4	An IO-ring or ESD defect would be unmeasured	Board: current-limited supplies; diode-check every pin (P0); ask imec to run both	REG:464-477
P1, N3, N7	IR drop not in timing (setup +15 ps at SS/1.08 V); single RC corner; hold at FF only	Slow hot parts may miss 100 MHz; fast parts could fail hold, which lowering f does not fix	Board: VDD 1.20 V, never 1.32 V; budget about 95 MHz; shmoo	REG:395-405, 445-456, 479-485
-	CPU1 app overwrites IPC slot 1	CPU0 boot delayed (its wait is bounded)	Do not write slot 1 before CPU0 boots	docs/debug/BUGREPORT_IPC_XIP_HANDSHAKE_CLOBBER_2026_08_20.md :3-5 PLAN:335-336
-	A spinlock page read acquires the lock; TideLink read-to-clear registers	A register sweep corrupts state	Blocklist (HZ-7, HZ-8)	
-	Perf-probe counters shadowed and saturating	Stale or 0xFFFFFFFF readings	SNAP before reading	PLAN:275-280
-	Simulation eth ROM (smoke_remap) $\neq$ silicon ROM: 172 of 320 words differ	FPGA-derived expected values are wrong	Use the class-G values in PROC §8.4	PROC:123; docs/tapeout/44-eth-rom-sim-divergence.md
-	Flash-boot result words collide: chip_core_flashmark (CPU1) writes 0x2D000000/04, the same words the hostio_fw CPU0 images use	One image's signature overwrites the other's	Give the CPU1 image a signature outside 0x2D000000--0x2D00004F and 0x2D001FF0	NMS/firmware/apps/chip_core_flashmark/main.c : 37-52; HFW:74-94
-	ptp_slave hard-codes NS_INCR = 40 and reports over UART; MDIO divider hard-coded for 25 MHz	Wrong rates on the ASIC	Rebuild for the board CLK; port to DMEM telemetry	research report 02 §3, §5

9 Characterisation plan (P15)

Aim. Build a pass/fail map of each die over frequency and core voltage, and separate core-timing failures from the QSPI boot-path risk (B2).

Axes.

- **Frequency:** $f = 10^9/N$ for $N = 20$ down to 10, i.e. 50.0, 52.6, 55.6, 58.8, 62.5, 66.7, 71.4, 76.9, 83.3, 90.9 and 100.0 MHz. Each keeps NS_INCR an integer. The oscillator's 25 MHz option ($N = 40$) is an optional extra slow point.
- **VDD:** 1.20 V down to 1.00 V in 25 mV steps (9 points). Do not go above 1.20 V: hold is only safe up to nominal REG:403-405.
- **Temperature:** room temperature; add hot and cold only if a chamber is available (**NOT FOUND**). Log a thermocouple near the glob top.

Order. Start at the known-good point (50 MHz, 1.20 V). Raise f at 1.20 V until the first failure plus one step. Then, at each f , lower VDD until the first failure plus one step. Never skip the S1 power sequence.

At each point, re-run:

Check	What	Failure points to
Power cycle, S1	NRST released with CLK stopped; record I(VDD), I(VDDIO), energy	–
P4 ladder	Rungs 0–6 and ROM fingerprint	CPU1 core or ROM path
P5.1 / P5.2	f_{HCLK} over 15 s; PHC rate with NS_INCR = N	Clock or PHC
P11 signatures	Flash boot of both CPUs	QSPI path (B2) if P4 and P5 pass
soc_char	App result block in the DMEM mailbox at 0x180003E0, read over HOSTIO	Core logic and memories at speed
P13.5 (selected points)	100 pings	RMII timing

Separating B2 from core timing. If flash boot fails at a given f but P4 and P5 pass, re-test that point with CLK_DIV = 2 via REG §3.1 workaround 2 or the CPU1 shim. If it then passes, the limit is the QSPI boot path, not the core.

Reading hold failures. A failure that appears at every frequency at a given VDD, including 50 MHz, points to hold, not setup. Lowering f does not fix hold REG:451-456.

Output. One CSV per die (point, pass/fail per check, currents, temperature) and one shmoo plot per die.

Time (this plan's estimate): $11 \times 9 = 99$ points at about 3 minutes each is about 5 hours per die per temperature, with the harness (Section 7, item 18).

10 Risks and open questions

#	Question or risk	Why it matters	Who can answer	Needed by
1	Closed 2026-10: 3.3 V (B-2). Was: VDDIO: 3.3 V or 2.5 V? rc4 STA views and the UPF use 3.3 V IO. docs/tapeout/10-tapeout-submission.md: 163 and ASIC/eth-chiplet/config/design_config.tcl:202 say 2.5 V; PROC marks it OPEN	Sets the VDDIO regulator, the flash part (SST26VF064B is 2.7–3.6 V only) and whether the 3.3 V RP2350 host needs level shifting	Design team; imec	Board order
2	POC behaviour and power sequencing NOT FOUND	Sequence S1 is a conservative guess	imec / TSMC POC datasheet	First power
3	Sawn die size, thickness, backside/substrate connection NOT FOUND	Finger distance ($\geq 1.5 \times$ thickness), die-attach pad potential, QFN $\leq 300 \mu\text{m}$	imec	Footprint freeze
4	Bonder sign-off: Zepler's wire (ball or wedge only?), finish and finger rules, stage size for the die card; staggered $58 \times 66 \mu\text{m}$ openings (P60 rules never run)	Gates the die-card layout and its finish; the motherboard is not affected	Zepler Cleanrooms	Card layout start
5	Expected SWD DPIDR value: candidate 0x6BA02477 (FPGA builds)	P8 cannot pass or fail against a reference	Design team; Arm SoC-400 docs	P8

#	Question or risk	Why it matters	Who can answer	Needed by
6	Closed 2026-10: no alternate; exactly SST26VF064B (B-3). Was: how does the alternate (MX25R6435F) handle 0x98 (ULBPR on SST26) and power-up block protection, given the boot-counter write?	A protected part blocks the counter write; unknown opcode behaviour	Datasheets	Board order
7	CLK part: programmable versus two fixed oscillators; clean start after OE	N1 mitigation depends on a glitch-free start	PCB designer	Board order
8	ERC and PadShorts never run on rc4	An IO-ring or ESD defect is unmeasured; P0 carries the risk	imec (on the GDS they hold)	First power
9	HOSTIO 330 Ω/5 kΩ values were measured on an FPGA IOBUF, not the ASIC pad	Levels need a re-check against the ASIC pad's datasheet values	Design team	Board order
10	Closed 2026-10: bond them, with DNP loopback footprints (B-1). Was: bond TL_TX[7:0]/TL_CLK_TX, or leave them unbonded?	Loopback continuity versus left-edge fan-out (Table 2)	Bring-up lead + bonder	Footprint freeze
11	HOSTIO role mapping is the HAPS-SX board's convention	Swapped IOACK/IODATA gives a dead link that looks like dead silicon	Schematic review	Schematic
12	Does uninitialised SRAM read as zero? (one-shot)	Decides HIO-107's replacement	First power	P4
13	CPU1 boot without flash: the docs disagree	P10 feasibility	PROC §6.3.1 four-command test	P10
14	TPS62869 power-up default output voltage (variant-dependent)	Must not exceed 1.20 V before the host clamps it	Datasheet	Board order
15	QSPI at SCLK = CLK/2 is uncharacterised (B2)	Flash boot may fail at high CLK	Bench (P9, P15)	–
16	SWD has never answered on a die. It has answered on FPGA (DPIDR 0x6BA02477, HAPS-SX 2026-09-09, KR260 2026-10-02)	Route B is more likely to open than first assumed; Route A must still work	Bench (P8)	–
17	Europractice packaging lead time, ENEPIG cost and lead time, PHY stock: not confirmed	Schedule of the packaged hedge and of the PCB	Europractice; PCBWay; distributors	Order time
18	The newest authority (REG) is untracked in the repository at 914c9e4	The citations in this plan could lose their target	Repo owner	Before this plan is published
19	The connector pinout is not frozen. Contract v1 fixes groups and counts, not positions; the pin-by-pin HSEC8 assignment is the next deliverable	Blocks the motherboard, die-card and K26-carrier layouts	Bring-up lead + PCB designer	Before any layout
20	Ground returns through the connector for the 15 always-driven 16 mA outputs: how many, and where. v1 says at least one GND per 2 signals	Ground bounce and crosstalk at the slot on QSPI, RMII TX and TL_TX edges	PCB designer	Pinout freeze
21	Kelvin sense through the connector: can the chosen regulators regulate on VDD_CORE_SENSE_P/N, or is sense for measurement only?	The shmoo must know VDD at the die, not at the regulator	PCB designer; datasheets	Schematic
22	Card thickness versus HSEC8: the connector needs a 1.60 mm card; confirm the bonder's stage and thickness limit, and ENIG + gold edge (or ENEPIG + hard gold, unconfirmed) on that card	A card that cannot be bonded or mated	Zepler; Eurocircuits / PCBWay	Card layout
23	K26 card IO isolation: the bus switches (loff-rated, e.g. SN74CB3Q3245) must hold all 48 slot nets clear of FPGA configuration pull-ups until CARD_READY, including while VIN_AUX ramps	Contention with motherboard pulls; a spurious IOACK low or QSPI traffic in Phase M	K26-carrier designer (FPGA_DAUGHTER_CARD_OPTIONS.pdf)	K26 carrier schematic
24	K26 card power through the connector: up to about 4 A at 5 V on VIN_AUX. The ~2.8 A/pin HSEC8 figure is a search snippet; ≥ 6 contacts, derated, plus the eFuse limit and the matching GND return	Contact heating, VIN_AUX droop at the SOM, eFuse nuisance trips	PCB designer; Samtec datasheet	Pinout freeze
25	Mechanical support for the K26 card (up to about 100 × 90 mm with heatsink and fan) in a card-edge slot; keep-out volume; same keying for every card	A heavy card levers the HSEC8 contacts and the motherboard	PCB / mechanical designer	Motherboard layout

#	Question or risk	Why it matters	Who can answer	Needed by
26	K26I versus K26C bitstream: the KR260 build targets the commercial part; the K26I needs a rebuild for xck26-sfvc784-2LV-1 (temperature grade only, per DS987)	A bitstream built for the wrong part will not load	FPGA developer (lab KR260s)	Phase M

A Memory map and master reachability

Region	Address	Size	Notes	Source
eth ROM / IMEM / DMEM	0x08000000 / 0x10000000 / 0x18000000	2K / 32K / 16K	ROM at 0x0 before REMAP	PLAN:247-252
CPU0-local: scratch RX, TX	0x30000000, 0x38000000	-	CPU0 and DAP only	AMAP:190-192
Ethernet MAC	0x40000000	-	CPU0 and DAP only	REG:213
CRC checker	0x48000000	-	CPU0 and DAP only	PROC:1153
eth APB: timer, UART, sysctrl	0x50000000 (+0, +0x1000, REMAP +0x2000)	-	REMAP decoded by the DAP only	REG:259
eth perf probe	0x60000000	-	CPU0 and DAP only	PROC:1153
DMA-250 config	0x20000000 (IIDR +0xFC8)	-		PLAN:230
QSPI registers	0x21000000	-	CTRL +0 (bit 8 XIP_ACTIVE); command +0x08/+0x10; AHB_SETUP +0x30; CLK_DIV +0x34	REG:172; PROC:86-97
PHC	0x22000000	-	CTRL +0, NS_INCR +0x08	PLAN:232
IPC mailbox	0x23000000	-	slot-1 data +0x10	PLAN:233
QSPI XiP	0x24000000	4 MB phys.	Blocklisted for host access	PLAN:234
CPU1 peripherals	0x28000000	-	timers, dual timer, UART +0x6000, watchdog +0x8000	AMAP:134-138
Boot gates / REMAP	0x29000000	-	bit 0 CPU1 REMAP (W1S); bit 1 inert (never set); bit 2 CPU0 release (W1S)	PLAN:236; PROC:1144
reset_ctrl	0x2A000000	-	+0x20: bit 0 CPU0_SWRST, bit 1 CPU1_SWRST	PLAN:237, 365
Event routing; debug group	0x2B000000; 0x2C000000	-	ETC, spinlock, perf probe, bus-fault monitor	PLAN:238-269
Shared SRAM	0x2D000000	8 KB	hostio_fw ABI at 0x2D000000 - -4F, 0x2D001FF0	PLAN:240; HFW:74-94
D2D local / peer	0x2E000000 / 0x2F000000	-	ROLE_CFG 0x2E032080; status 0x2E032084; peer blocklisted	PLAN:284-292
CPU1 ROM / IMEM / DMEM (admin view)	0x80000000 / 0x90000000 / 0x98000000	2K / 16K / 8K		PLAN:256-260
CPU0 / CPU1 debug windows	0xA0000000 / 0xB0000000	-	DAP only	PLAN:301

Master	Reaches	Source
HOSTIO / ADP (debug_m)	0x00-0x1F, 0x20-0x2F, 0x80-0x9F (top byte). Not the MAC, 0x5xxxxxxx or the debug windows, so it cannot halt a core	PLAN:225-301
SWD (one DAP: AP0 = CPU0, AP1 = CPU1)	Everything, including 0x5xxxxxxx and 0xA0/0xB0	REG:259
CPU0	Its local map including the MAC, plus 0x20-0x2F except 0x28	research report 02 §1
CPU1	Its local ROM/IMEM/DMEM plus 0x20-0x2F including 0x28. No MAC	REG:259
DMA-250	Both cores' memories, XiP (read-only), D2D. Hazardous across ports (HSEL)	research report 02 §1
DMA-230	Not on the chip	build/chip/flist/soc.flist
FPGA PS backdoor (eth_ss_0)	Tied idle on silicon	CB:282-290

B Boot flow and flash layout

CPU1 stage-0 ROM (stage0_bootrom_chip_core)

1. Writes the marker 0x0F1C0DE1 to its DMEM (0x98000200 from HOSTIO).
2. In APB command mode: WREN + ULBPR, then fast-reads 16 bytes of the counter sector (0x20000) and programs the first 0xFF byte to 0x00. It then polls the busy bit, bounded at 2^{24} polls.
3. Sets AHB_SETUP = 0x800B (fast read 0x0B, 8 dummy clocks, single SPI, 3-byte address) and XIP_ACTIVE. It halts silently if XIP_ACTIVE does not latch. CLK_DIV is never written.
4. Publishes 0xD15C0001 to IPC slot 1 (0x23000010), then reads the boot-attempt count.
5. Picks the valid, higher-seq table of TABLE0 (0x00000) and TABLE1 (0x10000). Candidate order: count < 2 gives [primary, other, golden]; count $\geq$ 2 gives [other, golden, primary].
6. Copies the first candidate's CPU1 image (entry 1) from XiP into IMEM and checks its CRC. On a pass it writes 0x4 (releases CPU0), then 0x1 (0x29000000 reads 0x5), and jumps. If every candidate fails it still writes 0x4, then halts in wfi.

Sources: ROM1:255-408; REG:170-196, 227-253.

CPU0 stage-0 ROM (stage0_bootrom)

1. Waits for 0xD15C0001. The wait is bounded at 0x40000 loops; it proceeds either way.
2. Halts if XIP_ACTIVE is clear. It never reprograms the controller.
3. Loads TABLE0 entry 0, or else the GOLDEN table's entry 0; it never reads TABLE1. The table must hold 2 or 3 entries. It copies the image over IMEM 0x10000000 and checks its CRC. If both fail it halts, writing 'X' to a UART that has no pad.
4. Sets VTOR, writes REMAP 0x50002000 = 1, and jumps.

Sources: ROM0:120-250. The whole chain was proven in gate-level simulation on the routed fp1505 netlist, not on rc4 (docs/tapeout/57-cpu0-boot-on-the-routed-netlist.md:151-167).

Flash layout

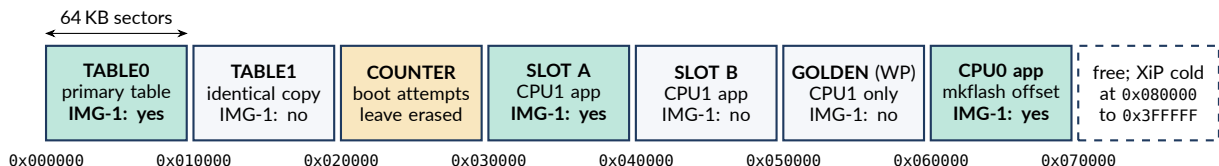


Figure 4. QSPI flash layout (offsets within the flash; XiP base 0x24000000). Green: sectors that IMG-1 (TABLE0-only) writes.

Sources: AMAP:376-381; MKF:120-143.

Table format (v2, little-endian). The 32-byte header holds magic 0x424F4F54 ("BOOT") at +0x00, version 2 at +0x04, num_entries at +0x08, seq at +0x0C, table_crc at +0x10, active_note at +0x14, and reserved words. Each 32-byte entry holds stage1_offset, stage1_size, app_offset, app_size, flags (bit 0 valid), stage1_crc, app_crc and a reserved word. table_crc is CRC32-IEEE (reflected, as binascii.crc32) over table bytes [0x14, 0x20 + $n \times 0x20$) AMAP:393-438.

Boot-counter policy. Each ROM boot programs one 0xFF→0x00 byte. A count $\geq$ 2 demotes the primary table. Confirmation means CPU1 publishes the boot-confirm token and CPU0, which owns the flash, erases the counter sector NMS/firmware/include/nanosoc_boot_confirm.h:8-26. With TABLE0 only, the demoted primary is still tried last, so the image keeps booting.

C Full pad list with bond-opening coordinates

Generated from `csv` by `figures/gen_figs.py`. Coordinates are bond-opening centres in μm , origin lower-left. “Design” is the $1600 \times 2000 \mu\text{m}$ frame; “seal ring” adds $20 \mu\text{m}$, giving the $1640 \times 2040 \mu\text{m}$ frame. Row: o = outer (PAD70GU_SL), i = inner (PAD70NU_SL). Openings are $58 \mu\text{m}$ along the edge by $66 \mu\text{m}$ deep. Order: left edge bottom to top, top edge left to right, right edge bottom to top, bottom edge left to right.

#	Signal	Instance (BuPAD_/uPAD_)	Edge	Row	IO cell	Design frame		Seal-ring frame	
						x	y	x	y
1	TL_RX0	TL_RX_0	left	o	PDDW16DGZ_G	36.8	162.5	56.8	182.5
2	TL_RX1	TL_RX_1	left	i	PDDW16DGZ_G	138.8	229.5	158.8	249.5
3	TL_RX2	TL_RX_2	left	o	PDDW16DGZ_G	36.8	296.5	56.8	316.5
4	TL_RX3	TL_RX_3	left	i	PDDW16DGZ_G	138.8	363.5	158.8	383.5
5	VDDIO	VDDIO_L_0	left	o	PVDD2DGZ_G	36.8	430.5	56.8	450.5
6	VSSIO	VSSIO_L_0	left	i	PVSS2DGZ_G	138.8	497.5	158.8	517.5
7	TL_CLK_RX	TL_CLK_RX	left	o	PDDW16DGZ_G	36.8	564.5	56.8	584.5
8	TL_RX4	TL_RX_4	left	i	PDDW16DGZ_G	138.8	631.5	158.8	651.5
9	TL_RX5	TL_RX_5	left	o	PDDW16DGZ_G	36.8	698.5	56.8	718.5
10	TL_RX6	TL_RX_6	left	i	PDDW16DGZ_G	138.8	765.5	158.8	785.5
11	TL_RX7	TL_RX_7	left	o	PDDW16DGZ_G	36.8	832.5	56.8	852.5
12	I2C_SCL	I2C_SCL	left	i	PDUW16DGZ_G	138.8	899.5	158.8	919.5
13	I2C_SDA	I2C_SDA	left	o	PDUW16DGZ_G	36.8	966.5	56.8	986.5
14	VDDIO	VDDIO_L_1	left	i	PVDD2DGZ_G	138.8	1033.5	158.8	1053.5
15	VSSIO	VSSIO_L_1	left	o	PVSS2DGZ_G	36.8	1100.5	56.8	1120.5
16	TL_TX0	TL_TX_0	left	i	PDDW16DGZ_G	138.8	1167.5	158.8	1187.5
17	TL_TX1	TL_TX_1	left	o	PDDW16DGZ_G	36.8	1234.5	56.8	1254.5
18	TL_TX2	TL_TX_2	left	i	PDDW16DGZ_G	138.8	1301.5	158.8	1321.5
19	TL_TX3	TL_TX_3	left	o	PDDW16DGZ_G	36.8	1368.5	56.8	1388.5
20	VDDIO	VDDIO_L_2	left	i	PVDD2DGZ_G	138.8	1435.5	158.8	1455.5
21	VSSIO	VSSIO_L_2	left	o	PVSS2DGZ_G	36.8	1502.5	56.8	1522.5
22	TL_CLK_TX	TL_CLK_TX	left	i	PDDW16DGZ_G	138.8	1569.5	158.8	1589.5
23	TL_TX4	TL_TX_4	left	o	PDDW16DGZ_G	36.8	1636.5	56.8	1656.5
24	TL_TX5	TL_TX_5	left	i	PDDW16DGZ_G	138.8	1703.5	158.8	1723.5
25	TL_TX6	TL_TX_6	left	o	PDDW16DGZ_G	36.8	1770.5	56.8	1790.5
26	TL_TX7	TL_TX_7	left	i	PDDW16DGZ_G	138.8	1837.5	158.8	1857.5
27	VDDIO (POC)	VDDIO_T_0	top	o	PVDD2POC_G	162.5	1963.2	182.5	1983.2
28	VDD	VDD_T_0	top	i	PVDD1DGZ_G	242.5	1861.2	262.5	1881.2
29	VSS	VSS_T_0	top	o	PVSS1DGZ_G	322.5	1963.2	342.5	1983.2
30	NRST	NRST_I	top	i	PDDW04DGZ_G	402.5	1861.2	422.5	1881.2
31	SWDCK	SWDCK_I	top	o	PDDW04DGZ_G	482.5	1963.2	502.5	1983.2
32	CLK	CLK_I	top	i	PDDW04DGZ_G	562.5	1861.2	582.5	1881.2
33	TEST	TEST_I	top	o	PDDW04DGZ_G	642.5	1963.2	662.5	1983.2
34	SE	SE_I	top	i	PDDW04DGZ_G	722.5	1861.2	742.5	1881.2
35	VDD	VDD_T_1	top	o	PVDD1DGZ_G	802.5	1963.2	822.5	1983.2
36	SWDIO	SWDIO_I0	top	i	PDUW08DGZ_G	882.5	1861.2	902.5	1881.2
37	VSS	VSS_T_1	top	o	PVSS1DGZ_G	962.5	1963.2	982.5	1983.2
38	VDDIO	VDDIO_T_1	top	i	PVDD2DGZ_G	1042.5	1861.2	1062.5	1881.2
39	VDDIO	VDDIO_T_2	top	o	PVDD2DGZ_G	1122.5	1963.2	1142.5	1983.2
40	VSSIO	VSSIO_T_0	top	i	PVSS2DGZ_G	1202.5	1861.2	1222.5	1881.2
41	VSSIO	VSSIO_T_1	top	o	PVSS2DGZ_G	1282.5	1963.2	1302.5	1983.2
42	VDD	VDD_T_2	top	i	PVDD1DGZ_G	1362.5	1861.2	1382.5	1881.2
43	VSSIO	VSSIO_T_2	top	o	PVSS2DGZ_G	1442.5	1963.2	1462.5	1983.2
44	VDDIO	VDDIO_R_0	right	o	PVDD2DGZ_G	1563.2	162.5	1583.2	182.5
45	VSSIO	VSSIO_R_0	right	i	PVSS2DGZ_G	1461.2	242.5	1481.2	262.5
46	RMII_MDC	RMII_MDC	right	o	PDDW16DGZ_G	1563.2	322.5	1583.2	342.5
47	RMII_REF_CLK	RMII_REF_CLK	right	i	PDDW04DGZ_G	1461.2	402.5	1481.2	422.5
48	RMII_RXD0	RMII_RXD0	right	o	PDDW04DGZ_G	1563.2	482.5	1583.2	502.5
49	VDDIO	VDDIO_R_1	right	i	PVDD2DGZ_G	1461.2	562.5	1481.2	582.5
50	VSSIO	VSSIO_R_1	right	o	PVSS2DGZ_G	1563.2	642.5	1583.2	662.5
51	RMII_RXD1	RMII_RXD1	right	i	PDDW04DGZ_G	1461.2	722.5	1481.2	742.5
52	RMII_TXD0	RMII_TXD0	right	o	PDDW16DGZ_G	1563.2	802.5	1583.2	822.5
53	RMII_TXD1	RMII_TXD1	right	i	PDDW16DGZ_G	1461.2	882.5	1481.2	902.5
54	RMII_TX_EN	RMII_TX_EN	right	o	PDDW16DGZ_G	1563.2	962.5	1583.2	982.5
55	RMII_MDIO	RMII_MDIO	right	i	PDDW16DGZ_G	1461.2	1042.5	1481.2	1062.5
56	RMII_CRD_DV	RMII_CRD_DV	right	o	PDDW04DGZ_G	1563.2	1122.5	1583.2	1142.5
57	VDDIO	VDDIO_R_2	right	i	PVDD2DGZ_G	1461.2	1202.5	1481.2	1222.5
58	VSSIO	VSSIO_R_2	right	o	PVSS2DGZ_G	1563.2	1282.5	1583.2	1302.5
59	HOSTIO0	HOSTIO_0	right	i	PDDW16DGZ_G	1461.2	1362.5	1481.2	1382.5

#	Signal	Instance	Edge	Row	IO cell	x	y	x	y
60	HOSTIO1	HOST_IO_1	right	o	PDDW16DGZ_G	1563.2	1442.5	1583.2	1462.5
61	HOSTIO2	HOST_IO_2	right	i	PDDW16DGZ_G	1461.2	1522.5	1481.2	1542.5
62	HOSTIO3	HOST_IO_3	right	o	PDDW16DGZ_G	1563.2	1602.5	1583.2	1622.5
63	HOSTIO4	HOST_IO_4	right	i	PDDW16DGZ_G	1461.2	1682.5	1481.2	1702.5
64	HOSTIO5	HOST_IO_5	right	o	PDDW16DGZ_G	1563.2	1762.5	1583.2	1782.5
65	HOSTIO6 ^a	HOST_IO_6	right	i	PDDW16DGZ_G	1461.2	1842.5	1481.2	1862.5
66	VDDIO	VDDIO_B_0	bottom	o	PVDD2DGZ_G	162.5	36.8	182.5	56.8
67	VSSIO	VSSIO_B_0	bottom	i	PVSS2DGZ_G	242.5	138.8	262.5	158.8
68	VDD	VDD_B_1	bottom	o	PVDD1DGZ_G	322.5	36.8	342.5	56.8
69	VSS	VSS_B_0	bottom	i	PVSS1DGZ_G	402.5	138.8	422.5	158.8
70	QSPI_IO0	QSPI_IO_0	bottom	o	PDDW16DGZ_G	482.5	36.8	502.5	56.8
71	QSPI_IO1	QSPI_IO_1	bottom	i	PDDW16DGZ_G	562.5	138.8	582.5	158.8
72	QSPI_IO2	QSPI_IO_2	bottom	o	PDDW16DGZ_G	642.5	36.8	662.5	56.8
73	VDDIO	VDDIO_B_1	bottom	i	PVDD2DGZ_G	722.5	138.8	742.5	158.8
74	VSSIO	VSSIO_B_1	bottom	o	PVSS2DGZ_G	802.5	36.8	822.5	56.8
75	QSPI_IO3	QSPI_IO_3	bottom	i	PDDW16DGZ_G	882.5	138.8	902.5	158.8
76	QSPI_SCLK	QSPI_SCLK	bottom	o	PDDW16DGZ_G	962.5	36.8	982.5	56.8
77	QSPI_nCS	QSPI_nCS	bottom	i	PDDW16DGZ_G	1042.5	138.8	1062.5	158.8
78	VDD	VDD_B_0	bottom	o	PVDD1DGZ_G	1122.5	36.8	1142.5	56.8
79	VSSIO	VSSIO_B_2	bottom	i	PVSS2DGZ_G	1202.5	138.8	1222.5	158.8
80	VDD	VDD_B_2	bottom	o	PVDD1DGZ_G	1282.5	36.8	1302.5	56.8
81	VSS	VSS_B_1	bottom	i	PVSS1DGZ_G	1362.5	138.8	1382.5	158.8
82	VDDIO	VDDIO_B_2	bottom	o	PVDD2DGZ_G	1442.5	36.8	1462.5	56.8

^a The CSV labels this bond pad edge=top; it is on the right edge (IO cell row 95 and x = 1513 μm).

D Address blocklist

The host tools must enforce this list. HZ numbers are PROC's hazard IDs.

#	Address / range	Rule	Why	Source
1	0x24000000 - 0x24FFFFFF (QSPI XiP)	Never access from HOSTIO or SWD	HZ-6. With no flash the wrapper stalls for 65536/f; with flash, a CPU executing from XiP can livelock and the watchdog never fires	PROC:1138; REG:204
2	0x2E0321A0 - 0x2E0321B8 (whole band)	Never access	HZ-1. Uninterruptible AXI hang; power cycle to recover. 0x2E0321AC/B0/B4 also hard-stall the CPU	PROC:1139; REG:160
3	0x2F000000 - 0x2FFFFFFF (peer aperture)	Never access: there is no peer die	HZ-2. No default responder. A failed read parks the port and the next access wedges the host	PROC:1140
4	0x2E032080 ROLE_CFG	Never write, except in P2b with the B-1 loopback fitted	Starts TideLink TX clock and data toggling at the CLK rate; RX never calibrates without the loopback	REG:78-98
5	0x2E032020, 024, 038, 15C, 160, 168	Never read	HZ-8. Read-to-clear and auto-increment: a read is a write	PROC:1142
6	0x2C100000, 0x2C100100, 0x2C100200	Never read in a sweep. Use the DBG page 0x2C100200 only on purpose	HZ-7. A read acquires the lock, and both cores are live	PROC:591, 1141
7	0x2E000000 - 0x2E003FFF	No writes	HZ-3 (while the link is up). Keep the habit on this board	PROC:1115
8	0x29000000 bit 1	Never set	Inert but write-1-set and permanent	PROC:1144
9	0x29000000 bit 0	Only in P10, on purpose	HZ-4. Irreversible until power-on	PROC:719
10	0x2A000020 = 2 (CPU1_SWRST)	Expect the link to drop	HZ-5. Resets the whole fabric including HOSTIO	PROC:848
11	Flash erase or program via 0x21000000	Not on the first die	The only HOSTIO route to CPU0 firmware; a state the ROM cannot parse loses the boot	PROC:1143
12	Apps default_slave_error_walk, busfault_irq_route	Never run	They poke 0x2E/0x2F believing them unmapped	REG:159-161
13	HIO-203/204/208 (Tier 2)	Never before firmware-dependent tests in the same power cycle	They wipe eth IMEM and shared SRAM	PROC:1147

#	Address / range	Rule	Why	Source
14	Two-master rule	Never two of {DMA-250, DAP, ADP} on CPU1's or the eth subsystem's memories at once	HSEL defect in rc4	research report 01 \$7
15	DMA-250 with any address in 0x2E000000 - -0x2FFFFFFF	Never (source or destination)	Erratum D2D-DMA: silent corruption, DONE with no error	docs/chip-reference-manual/review/closure/item11-tl053.md

E Source references

Repository paths are relative to the nanosoc-ethernet-chiplet root, as read at 914c9e4.

Alias	Path	Note
REG	docs/tapeout/67-rc4-issue-register.md	rc4 issue register, 2026-09-23; newest authority. Untracked at 914c9e4
PROC	docs/bringup/ASIC_FIRST_SILICON_HOSTIO_PROCEDURE.md	HOSTIO first-silicon procedure; \$0.3 superseded by REG
PLAN	docs/bringup/HOSTIO4_FUNCTIONAL_TEST_PLAN.md	HOSTIO functional test plan (memory map)
BSC	docs/bscan/BRINGUP.md	Boundary-scan bring-up
PADS	ASIC/tech_wrappers/tsmc65/nanosoc_eth_chiplet_pads.v	Pad ring source; the rc4 netlist agrees
PNR	ASIC/eth-chiplet/build/rc4-20260829/outputs/nanosoc_eth_chiplet_pads_rc4_pnr.v	Shipping netlist
FP	ASIC/eth-chiplet/build/rc4-20260829/db_rc4/nanosoc_eth_chiplet_pads.fp.gz	rc4 floorplan
CSV	ASIC/submissions/padring_as_built_rzg_20260823/nanosoc_eth_chiplet_pads_as_built_pads_only.csv	Matches the rc4 floorplan on 168/168 rows
PTAB	src/rtl/bscan/pad_table.json	Pad REN/OEN ties, TAP pins, IDCODE
SDC	ASIC/eth-chiplet/build/holdco-20260827/work/nanosoc_eth_chiplet_pads_routed_src/mmc/modes/default_constraint_mode/default_constraint_mode.sdc	The SDC rc4 was timed with
UPF	ASIC/genus-innovus/inputs/nanosoc_eth_chiplet_pads.upf	Supply ranges
CB	sys_desc/chip_boundary/nanosoc_eth_chiplet.yaml	Chip boundary
NMS	nanosoc-multicore-system/	Submodule at 2adcd34
AMAP	NMS/firmware/include/nanosoc_multicore_addrmap.h	Address map, flash format
ROM1	NMS/firmware/bootloader/stage0_bootrom_chip_core/main.c	CPU1 boot ROM source
ROM0	NMS/firmware/bootloader/stage0_bootrom/main.c	CPU0 boot ROM source
flash_pack.py	NMS/nanosoc_arch_tech/firmware/testcodes/bootloader/flash_pack.py	The v2 packer
MKF	ASIC/gls-netlist/flash/mkflash.sh	Wrapper; check_flash.py alongside
RIG	scripts/rig/eth_chiplet/	HOSTIO client, suite, hostio_fw
HFW	scripts/rig/eth_chiplet/hostio_fw/README.md	hostio_fw ABI and signatures (locally modified at 914c9e4)
PRMU	NMS/nanosoc_arch_tech/rtl/slcorem0p_tech/src/verilog/slcorem0p_prmu.v	Clock and reset
XDBG	docs/design/CROSS_DIE_DEBUG_PLAN.md	DAP and APs

Research reports (2026-09-30, inputs to this plan; they carry the full citation trail): 01 die and PCB electricals; 02 functionality, firmware and tests; 03 PCB tools, chip-on-board and parts; 04 die-card slot contract v0 (superseded); 05 flip-chip versus daughter card; 06 slot contract v1 (interchangeable die card / K26 FPGA card). Companion study: FPGA_DAUGHTER_CARD_OPTIONS.pdf (this directory).

External sources (from report 03, checked 2026-09-30):

- Würth Elektronik: wire-bonding design-rule poster v1.2, <https://www.we-online.com/files/pdf1/design-rules-wire-bonding-cbt-en.pdf>; webinar 108, <https://www.we-online.com/files/pdf1/presentation-webinar-108-wesystems-bonding-and-more-cbt-en.pdf>.
- Europactice ASIC prototype packaging design rules 2024, https://europactice-ic.com/wp-content/uploads/2020/06/ASIC_Prototype_Packaging_Design_Rules_2024.pdf; packaging service, <https://europactice-ic.com/services/packaging/asic-packaging/>.

- Zepler Cleanrooms back-end, <https://www.southampton.ac.uk/research/institutes-centres/zepler-cleanrooms/backend-dice-bond>.
- PCBWay gold wire-bonding finish, <https://www.pcbway.com/capabilities.html>; JLCPCB ENEPiG vs ENiG, <https://jlcpcb.com/blog/enepig-vs-enig-surface-finishes>.
- Samtec HSEC8 (<https://uk.rs-online.com/web/p/edge-connectors/1798778>); Samtec Q Strip; Eurocircuits HDI pool and gold-edge-connector option, <https://www.eurocircuits.com/services/hdi-pool/>; Technion TPT HB16 page (stage size, search snippet); arXiv 1112.1037 (Au stud bumps on single dies). AMD Kria K26 SOM data sheet DS987 v1.2 (tables 1, 22, 25), K24 data sheet DS985, K26 carrier design guide UG1091; Mouser UK stock and prices for SM-K26-XCL2GI and the K26C (2026-09-30); TI SN74CB3Q3245.
- Parts: LAN8720A DS00002165B; TPS62869, TPS7A94, INA228 and TS3A27518E (ti.com); MX25R6435F (DigiKey); J-Link BASE (segger.com).
- Tools: SKiDL <https://github.com/devbisme/skidl>; Diode <https://github.com/diodeinc/pcb>; KiCad CLI <https://docs.kicad.org/9.0/en/cli/cli.html>; Freerouting <https://github.com/freerouting/freerouting>; KiKit <https://github.com/yaqwsx/KiKit>.